

GAUGE AND GRAVITATIONAL RESPONSES OF A FIELD

SINA MONTAZERI
UNIVERSITY OF NORTH TEXAS

ABSTRACT. Conventional field theories specify fields, gauge groups, and actions by extrinsic assignment upon a pre-existing manifold. However, a tangent-normal bundle metric uniquely determines the curvature family, gauge invariance, and variational energy of a field. Metric compatibility across these bundle sectors generates the underlying spatial geometry, local coordinates, and differential calculus directly from the metric itself. Continuum geometry is therefore deduced directly from the Zermelo–Fraenkel axioms without postulating an antecedent background manifold.

The curvature of this field decomposes into tangential Gauss, mixed Codazzi, and normal Ricci sectors, identifying spatial Riemannian curvature, extrinsic kinematics, and non-Abelian gauge field strengths within a single connection. Across the carrier bundle over spatial three-manifolds, the maximal compact isometry group preserving unimodular metrics and polar complex structures uniquely derives the exact Standard Model gauge group $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)]/\mathbb{Z}_6$, with electric charge and flux quantization forced by the canonical phase identity $J^2 = -I$, while higher-rank grand-gauge extensions are dynamically obstructed.

Dissipation of curvature energy drives an autonomous response flow on the configuration space of spatial metrics. Operating on spatial geometries prior to spacetime assembly, the configuration motion metric is strictly positive-definite and coercive, eliminating the indefinite DeWitt supermetric catastrophe. The chronological foliation of this trajectory generates physical four-dimensional spacetime, recovering the Lorentzian metric and relativistic light cone from the contact saturation of unilateral domain constraints, with gradient dissipation dictating the thermodynamic arrow of time, while Gauss–Codazzi–Ricci integrability enforces the non-linear Einstein–Yang–Mills initial-data constraints.

At the quantum level, canonical commutation relations and observable algebras descend directly from the geometric response derivation on cotangent phase space, prior to any axiomatic quantum postulate. The operator-ordering and non-closure anomalies of Dirac constraint quantization resolve through the maximal weak adjoint closure of the constraint core, yielding a strictly positive, self-adjoint Hamiltonian constraint operator whose spectral minimization governs physical quantum states on standard separable functional domains.

When autonomous curvature flow drives configuration geometry toward singular collapse, unilateral domain constraints arrest the contraction. Physical matter is the localized restoring strain generated at this threshold, identifying physical stress-energy directly with the reaction forces halting geometric degeneracy. Because the kinetic sector of the configuration metric carries a hyperbolic signature, this reactive coupling is quadratic, mathematically enforcing universal gravitational attraction and fixing Newton’s gravitational constant directly from response amplitudes.

This quadratic response forces strict negative normal curvature $K = -b_4^2 < 0$, activating McKean’s Riemannian spectral theorem on the physical spatial leaf, while unilateral dilatation scale-breaking at the matter spacing floor b_3 lifts all residual flat modes, generating an exact, strictly positive non-perturbative physical mass gap $m = |b_4|/2 > 0$ with spectral floor $\lambda_0 = b_4^2/4 > 0$.

1. INTRODUCTION

Gauge theory and gravitational initial-data geometry differ in what they take to be prior to variation. Gauge theory begins with an internal connection on a chosen base, while gravitational initial data begins with a metric and a second fundamental form, as in the ADM formulation [3]. Yet locality, internal directions, admissible configurations, and the governing law remain distinguishable structures in both descriptions. The problem investigated here is to determine whether these disparate structures must be supplied independently or whether they follow unconditionally from data recording local variation, internal organization, and their structural roles.

In the present work, the metric is established as the foundational equilibrium point. Tracing the internal consistency requirements of the metric-germ configuration space and its curvature response mathematically deduces the algebraic properties of the base metric, local manifold charts, bundle dimensions, metric signatures, the differential calculus, and the real continuum directly within Zermelo–Fraenkel set theory [137, 43, 68]. The construction does not rely on statistical continuum approximations, large- N matrix expansions, or discrete simplicial cutoffs. Smooth manifold geometry and its differential calculus emerge as exact consistency requirements of the metric equilibrium point. An exhaustive comparative analysis demonstrating how this construction resolves historical impasses in causal sets, matrix models, loop quantum gravity, string compactifications, and noncommutative geometry is presented in Section 2.

Ambient bundle curvature decomposes into tangential Gauss, mixed Codazzi, and normal Ricci sectors across the Serre–Swan carrier bundle, establishing that gauge fields and spatial geometry are the inseparable geometric projections of a single metric-compatible connection. The second variation of the curvature-energy functional generates a four-block weak Hessian whose off-diagonal blocks provide non-vanishing cross-coupling between gauge connections and shape perturbations. On constant-rank strata, polar decomposition resolves the coupled amplitudes into canonical symmetric and complex factors, where the phase operator induces an almost complex structure satisfying $J_X^2 = -I_H$. On spatial three-manifolds, the maximal compact group of automorphisms preserving unimodular carrier volume and this polar phase uniquely derives the exact Standard Model gauge group $[\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)]/\mathbb{Z}_6$, with charge and flux quantization forced by the phase algebra, while higher-rank grand-gauge extensions are dynamically obstructed by finite carrier rank and Hessian stability.

An autonomous first-order response flow governs the dynamics on configuration space without presupposing an external spacetime manifold or coordinate clocks. This flow generates an internal dynamical trajectory, establishing causal ordering prior to the introduction of any spacetime atlas. Spacetime is not an antecedent dimensional container upon which fields evolve. Physical four-dimensional spacetime is reconstructed as the historical foliation of metric-germ degrees of freedom along the autonomous dissipative gradient flow. The resulting spacetime geometry satisfies the Gauss–Codazzi–Ricci integrability conditions, recovering non-linear general relativity and the Arnowitt–Deser–Misner initial-data constraints on spatial slices, while the Lorentzian causal light cone and thermodynamic arrow of time emerge unconditionally from the contact saturation of unilateral domain constraints.

At the quantum and phase-space level, canonical commutation relations and observable algebras descend directly from the geometric response derivation on cotangent phase space without requiring an axiomatic quantum postulate. The Hamiltonian derivation of the signed Legendre function descends to a gauge-intertwining $*$ -derivation on the local canonical commutation relation algebra. Within a gauge-compatible representation, total symmetrization provides the unique finite-label quantization of real polynomials of degree at most four satisfying leading-symbol and linear symplectic-equivariance conditions, circumventing the Groenewold–van Hove obstruction on this domain. The operator-ordering and non-closure anomalies of Dirac constraint quantization are resolved by the maximal weak adjoint closure of the constraint core, yielding a strictly positive, self-adjoint Hamiltonian constraint operator whose spectral minimization governs physical quantum states on standard separable functional domains. The bounded observables generate unital C^* -algebras containing bounded functions of this constraint operator, curvature contractions, and charge-sector transitions.

Physical models demonstrate the geometric origin of mass, spectral isolation, and the nature of matter. Matter is not an independent substance placed upon geometry, but the localized reactive constraint strain tensor of saturated geometric configuration constraints under the autonomous variational flow. In unsourced Abelian vacuum, flat normal curvature permits massless propagation without an energy gap. When non-Abelian fields or matter sources are present, gauge currents couple quadratically to gravitation, mathematically enforcing universal gravitational attraction and fixing Newton’s gravitational constant directly from response amplitudes. This quadratic response forces strict negative normal curvature $K = -b_4^2 < 0$. By McKean’s Riemannian spectral theorem on complete negatively curved Riemannian manifolds [91], this negative curvature acts as an absolute geometric barrier, enforcing an exact spectral floor $\lambda_0 = |K|/4 = b_4^2/4$ and an invariant physical mass gap $m = |b_4|/2$ without invoking an extrinsic scalar field or tachyonic potential. Apparent zero-mode obstructions on flat three-tori are resolved on physical spatial slices, where Hodge–de Rham theory on S^3 [60, 30, 124] eliminates harmonic zero modes, and the unilateral scale-floor constraint $\varepsilon \geq b_3$ [70, 76] breaks dilatation invariance, lifting the radial mode into the positive spectrum.

The presentation proceeds in this order. Section 2 establishes the theoretical context, the comparative analysis against existing quantum gravity and gauge programs, and the axiomatic descent to Zermelo–Fraenkel set theory. Section 3 constructs the unimodular metric-germ configuration space and its Levi-Civita connection. Section 4 derives the geometric energy, four-block weak Hessian, and stratified polar calculus. Section 5 proves the curvature decomposition and the autonomous response flow on the configuration quotient. Section 6 reconstructs Lorentzian spacetime, the shape line, and the ADM initial-data identification. Section 7 establishes variational matter couplings, quadratic gravitation, and the geometric spacing floor. Section 8 derives the internal gauge groups, carrier stabilizers, and geometric obstructions to higher-rank group extensions. Section 9 derives the geometric Noether currents, local charge conservation, and interface current localization. Section 10 constructs operational observer frames, radar metric reconstruction, and deterministic extrinsic curvature inversion. Section 11 realizes concrete physical models, solitons, and the McKean curvature mass gap. Section 12 synthesizes the results and the resolution of geometric obstructions. The appendices

contain the detailed metric-germ quotient proofs, polar calculus, curvature expansions, spacetime reconstruction maps, operator algebras, and simulation tables in logical proof-dependency order.

2. RELATION TO EXISTING THEORIES AND PROGRAMS

The theoretical programs considered below differ in where they place the boundary between initial data and derived structure. Standard formulations begin with an extrinsic manifold, metric, connection, phase space, or algebra of observables already specified. Others attempt to recover geometry from discrete posets, matrix limits, or spectral triples. The present work establishes that the metric is the equilibrium point, from which local manifold charts, the Levi-Civita connection, configuration space, curvature response, Lorentzian spacetime history, matter ontology, and the mass spectrum are determined. The comparisons below are organized across seven successive stages of this argument flow.

2.1. Axiomatic Descent and Compatibility Requirements of the Base Metric. Several quantum gravity programs formulate kinematic variables prior to the introduction of a smooth manifold. Causal set theory replaces continuous spacetime with a locally finite partially ordered set, seeking to recover continuum Lorentzian geometry through Poisson sprinkling and statistical averaging [11]. Matrix models obtain geometry and gauge fluctuations from large- N matrix algebras [63, 109], or generate emergent spacetime from quantum renormalization group scaling [80, 81]. Quantum circuit proposals seeking space from abstract Hilbert space reconstruct spatial metrics from quantum mutual information matrices [20, 19], while the amplituhedron eliminates local spacetime and unitarity in favor of positive Grassmannian geometry [2]. In twistor theory, Penrose formulated null rays and complex projective geometry prior to real spacetime points [96]. Causal fermion systems formulate spacetime as the support of a minimizing measure on a set of linear operators [41], and causal dynamical triangulations generate four-dimensional spacetimes through path integrals over simplicial complexes [1].

Each of these programs confronts the difficult challenge of recovering smooth four-dimensional Lorentzian spacetime through statistical limits, large- N expansions, or discretization cutoffs. In minimal-length scenarios for quantum gravity [62], the existence of an invariant UV cutoff is introduced to regularize short-distance behavior. In contrast, descending from the metric equilibrium point provides conditional exact determination, where an invariant spacing floor is determined by the boundary of the constraint cone without being imposed as an external quantum modification.

The base metric h is not postulated as an unconstrained axiomatic choice. Tracing the consistency requirements of the total connection on $TM \oplus N$ and its curvature response mathematically forces the algebraic properties of the base pairing. First, the block structure of the total connection requires a bilinear form on TM . This linear requirement rules out direction-dependent Finsler structures $F(x, v)$ [25], whose metric tensors depend nonlinearly on tangent directions and fail to define a canonical linear adjoint. Second, the existence and uniqueness of the adjoint shape operator $K^\dagger: N \rightarrow TM$ in the metricity condition $\nabla(h \oplus g_N) = 0$ forces h to be nondegenerate. Third, the existence and uniqueness of a canonical torsion-free metric-compatible connection forces h to be symmetric. These three

consistency conditions establish the base metric pairing,

$$(1) \quad h: TM \times TM \rightarrow \mathbb{R}, \quad h(X, Y) = h(Y, X), \quad \det(h_{ij}) \neq 0.$$

The base metric h is therefore not an arbitrary background field, but the unique minimal bilinear structure admitting an adjoint shape operator and a torsion-free connection. On this pairing, Koszul's formula [74] provides the Koszul connection determination,

$$(2) \quad \begin{aligned} 2h(D_X Y, Z) &= X(h(Y, Z)) + Y(h(X, Z)) - Z(h(X, Y)) \\ &+ h([X, Y], Z) - h([X, Z], Y) - h([Y, Z], X). \end{aligned}$$

uniquely fixing the Levi-Civita connection $D = \text{LC}(h)$ directly from metric derivatives. Tracing these requirements further recovers local manifold charts, bundle dimensions, metric signatures, the differential calculus, and the real continuum, terminating at the axioms of Zermelo–Fraenkel set theory [137, 43, 68]. Within the smooth metric-germ branch, continuum geometry is derived without statistical approximations or discretization artifacts.

2.2. Configuration Space and Kinematic Decoupling. Canonical formulations of gravitation, beginning with Dirac's generalized Hamiltonian mechanics [32] and Wheeler's geometrodynamics [127, 31], take a spatial three-manifold and its Riemannian metric space as initial starting data. In relational mechanics, Barbour and Bertotti formulated dynamics directly on the conformal quotient of Riemannian metrics modulo spatial diffeomorphisms to eliminate external coordinate clocks [6]. Rigorous analysis of this configuration space was established by Ebin's slice theorem, proving that the space of Riemannian metrics modulo spatial diffeomorphisms forms a stratified infinite-dimensional manifold [34]. In the Arnowitt–Deser–Misner (ADM) formulation, a spatial Riemannian metric h and an extrinsic curvature tensor K form canonical initial data whose evolution is governed by the scalar Hamiltonian and vector momentum constraints [3]. In quantum geometrodynamics, these constraints are promoted to operator conditions on superspace, inheriting the DeWitt supermetric with indefinite signature $(-, +, +, +, +, +)$. This indefiniteness leads directly to the problem of time and impedes the construction of a positive-definite inner product on quantum states. In Euclidean quantum gravity, Gibbons and Hawking evaluated the gravitational partition function via path integrals over Riemannian metrics with boundary terms [46], but encountered the conformal factor instability—the gravitational Euclidean action is unbounded from below under conformal transformations, requiring an ad-hoc contour rotation into the complex plane.

In contrast, the metric-germ construction decouples the configuration motion metric from spacetime signature. Factoring out gauge redundancies on the quotient space $\mathcal{Q} = \mathcal{U}_0 / \ker \sigma$ isolates physical connection variations δA and extrinsic shape variations δK . Integrating these localized variations defines the positive configuration motion metric,

$$(3) \quad G((\delta A, \delta K), (\delta A, \delta K)) = \int_U \left[\frac{1}{2} |\delta A|_{h \otimes g_N}^2 + |\delta K|_{h \otimes g_N}^2 \right] d\mu_h > 0.$$

Because the kinetic pairing is an integrated L^2 sum of positive quadratic norms, G is strictly positive-definite across all non-zero variations. This separates the positive physical motion metric on configuration space from the reconstructed Lorentzian

signature of spacetime, resolving the indefiniteness that besets the DeWitt supermetric and eliminating the Euclidean conformal instability.

2.3. Metric Direct-Sum Bundles and Curvature Decomposition. The relationship between spacetime geometry and internal gauge connections has followed several historical lines. Weyl first attempted to synthesize gravitation and electromagnetism by introducing a conformal scale connection [125], a proposal obstructed by the non-integrability of atomic spectral clocks, before subsequently establishing that local $U(1)$ phase invariance provides the geometric foundation of electrodynamics [126]. Kaluza and Klein obtained Maxwell electrodynamics from a five-dimensional Riemannian metric carrying an isometric circle action [69, 72]. Following the formulation of non-Abelian gauge theory by Yang and Mills [135], Wu and Yang established the global differential-geometric dictionary identifying gauge fields with connections on principal fiber bundles [134]. In quantum systems, Wilczek and Zee demonstrated that non-Abelian gauge structures emerge dynamically from the adiabatic holonomy of degenerate sub-bundles without postulated gauge fields [131]. Concurrently, gauge theories of gravitation represented gravitational fields by Lorentz or Poincaré connections on an extrinsic manifold [119, 71, 105, 57, 4]. Larger-connection constructions placed gravitational spin connections and internal gauge connections into different sub-blocks of an enlarged connection for a larger Lie group [86, 93, 85], while pure connection gravity sought to recover the metric from internal connection curvature [75].

These programs either postulate additional unobserved spatial dimensions, treat the spacetime manifold as an independent background upon which gauge connections live, or pack disparate fields into a single Lie algebra with ad-hoc symmetry breaking potentials. In string and M-theory, gauge and gravitational interactions are jointly formulated in ten or eleven spacetime dimensions, requiring compactification on Ricci-flat Calabi–Yau manifolds [48, 18] or holographic dualities connecting bulk Anti-de Sitter gravity to boundary conformal fields, boundary entanglement entropy, and quantum error-correcting codes [88, 87, 121, 102, 110, 14, 94, 53, 35, 112]. These approaches require negative cosmological constants, auxiliary boundaries, and unobserved spatial dimensions. In contrast, the present construction distinguishes the tangent bundle TM and the internal bundle N as vector bundles over the same base manifold without extra dimensions, compactification moduli, or holographic extrapolation. Requiring metric compatibility $\nabla(h \oplus g_N) = 0$ on the direct sum $TM \oplus N$ uniquely fixes the off-diagonal blocks as the shape operator K and its adjoint $-K^\dagger$, establishing the assembled metric connection,

$$(4) \quad \mathcal{C} = \begin{pmatrix} D & -K^\dagger \\ K & A \end{pmatrix}, \quad \nabla(h \oplus g_N) = 0.$$

Evaluating the curvature two-form of $\mathcal{C}$ decomposes ambient geometry into tangent, mixed, and normal sectors, which define the curvature decomposition and energy functional,

$$(5) \quad F_{\mathcal{C}} = \begin{pmatrix} T & -M^\dagger \\ M & N \end{pmatrix}, \quad \Phi(\Psi) = \frac{1}{2} \int_U |F_{\mathcal{C}}|^2 d\mu_h.$$

The tangential Gauss sector T , mixed Codazzi sector M , and normal Ricci sector N determine spacetime gravitation, extrinsic kinematics, and Yang–Mills gauge strength simultaneously. The positive curvature-energy functional $\Phi(\Psi)$ provides

a bounded-from-below action without introducing ad hoc potential terms. Furthermore, the internal gauge group is not postulated by extrinsic group selection; the compact unitary stabilizer of the transverse complex structure is derived along the shape trajectory, aligning with Berger’s classification of Riemannian holonomy groups and metric stabilizers [9].

2.4. Algebraic Reconstruction Without A Priori Spectral Triples. Established reconstruction theorems recover geometric spaces from localized algebraic data. Gelfand and Naimark proved that every commutative C^* -algebra is isometrically $*$ -isomorphic to the algebra of continuous functions on its compact spectrum [45]. Serre and Swan established that finitely generated projective modules over $C^\infty(M)$ correspond bijectively to the smooth sections of vector bundles over M [106, 111]. Noncommutative geometry extends this duality through spectral triples (A, H, D) , reconstructing a Riemannian manifold from an abstract algebra, a Hilbert space, and a Dirac operator, while deriving the Einstein–Hilbert and Yang–Mills actions via the spectral action principle [28, 29, 22]. However, noncommutative geometry requires an extrinsic Hilbert space, a real structure J , and fine-tuned grading operators that are difficult to justify dynamically.

The present formulation utilizes Gelfand–Naimark [45] and Serre–Swan [106, 111] as local algebraic bridges without adopting noncommutative spectral triples. Restricting commutative observables $\mathcal{A}$ to local chart domains recovers the smooth base manifold M , while vector bundle sections correspond bijectively to module projections, establishing the local algebraic module bridge,

$$(6) \quad C^\infty(M) \cong C(\text{Spec}(\mathcal{A})), \quad \Gamma(TM \oplus N) \cong \mathcal{P} \cdot (C^\infty(M))^k.$$

The base metric determines the Levi-Civita connection directly via Koszul’s formula (2), and field dynamics is governed by the curvature functional $\Phi(\Psi)$ in (5), avoiding the need to postulate an antecedent Dirac operator at the initial stage.

2.5. Autonomous Configuration Dynamics and Constraint Minimization. Hypersurface-deformation kinematics governs the canonical evolution of geometry. The Hojman–Kuchař–Teitelboim (HKT) theorem establishes that any Hamiltonian theory on a geometrodynamical phase space satisfying the hypersurface deformation algebra is uniquely forced to be general relativity [61]. In perturbative quantum gravity, Feynman’s pioneering work [40] demonstrated that quantizing metric fluctuations via path integrals [39] produces gauge-redundancy anomalies at the loop level, requiring the introduction of ghost fields to restore unitarity. Following the proof of renormalizability for non-Abelian gauge theories [113, 114], ’t Hooft and Veltman established that while pure general relativity is one-loop finite on-shell, coupling gravitation to matter fields produces uncontrollable, non-renormalizable ultraviolet divergences [115]. At the non-perturbative level, Dirac constraint quantization of the hypersurface algebra is severely obstructed by structure functions that depend on the spatial metric, causing operator-ordering anomalies and non-closure in the quantum theory [32, 58]. In canonical loop quantum gravity and spin foam models, quantization is constructed on non-separable polymer Hilbert spaces using holonomies of the Ashtekar–Barbero connection [4, 5] and surface flux operators, yielding discrete spectra for geometric area and volume operators [101, 100, 118]. Furthermore, the Groenewold–van Hove theorem proves that no faithful Lie-algebra homomorphism exists from classical Poisson brackets to quantum commutators for polynomials beyond degree two [50, 120].

In the present setting, dynamics is governed autonomously by a first-order response flow on configuration space without presupposing a spacetime metric. While Hamilton’s Ricci flow establishes an intrinsic parabolic evolution on the space of Riemannian metrics driven by Ricci curvature [52], the present construction drives geometry by the negative gradient of the total curvature-energy functional with respect to the configuration metric, generating the autonomous configuration flow,

$$(7) \quad \frac{d\Psi}{dt} = -\nabla_G \Phi(\Psi) = \mathcal{V}(\Psi(t)).$$

Spacetime is constructed as the historical foliation along the distinguished shape line $q = 1$, satisfying the Gauss–Codazzi–Ricci integrability conditions and recovering the ADM constraints prior to assuming a hypersurface deformation representation. In thermodynamic approaches to gravitation, Jacobson demonstrated that Einstein’s field equations arise as a local horizon equation of state [66]. Similarly, in Sakharov’s induced gravity, spacetime elasticity and gravitational dynamics arise from the quantum fluctuations of underlying fields [103]. Non-linear general relativity emerges from the integrability and unilateral boundary equilibrium of the autonomous dissipative response flow. On cotangent phase space, this response descends to a gauge-intertwining $*$ -derivation on the local canonical commutation relation algebra, aligning with the net-of-algebras formulation of algebraic quantum field theory [51, 15] and Doplicher–Roberts gauge reconstruction [33]. Symmetrization uniquely quantizes polynomials of degree at most four satisfying linear symplectic equivariance, circumventing the Groenewold–van Hove obstruction on this domain. The maximal weak adjoint closure of the symmetric constraint generates the positive Master constraint operator,

$$(8) \quad \mathbf{M} = \mathbf{C}^\dagger \mathbf{C} \geq 0, \quad \ker \mathbf{M} = \ker \mathbf{C}.$$

implementing Thiemann’s constraint programme [117, 118] and transforming constraint non-closure into a positive spectral minimization problem on standard separable functional domains [13, 98].

2.6. Emergent Matter, Geometrodynamics, and Solitons. Historical attempts to derive physical matter directly from geometry rather than postulating extrinsic stress-energy tensors began with Wheeler’s geometrodynamics [128, 127]. In Wheeler’s “mass without mass” program, localized configurations of source-free classical electromagnetic and gravitational fields, termed geons, were proposed as classical models of stable particle-like matter. However, classical geons are generically unstable against gravitational collapse or radiative dispersion, and classical pure geometry fails to generate half-integer fermion spin or anti-commuting field algebras without inserting extrinsic spinor structures by hand.

Topological soliton programs, initiated by Skyrme [107, 108] and extended by Manton [89] and Witten [133], circumvented stability failures by introducing non-linear sigma models where baryons emerge as topologically stable spatial field knots with non-zero winding numbers $\pi_3(\text{SU}(2)) \cong \mathbb{Z}$, with fermionic statistics realized by the Wess–Zumino–Witten topological action. Nevertheless, topological soliton theories postulate an antecedent non-linear matter Lagrangian defined on an antecedent spacetime manifold. Matter is represented by continuous spatial configurations of an antecedent matter field rather than arising from the geometric incompressibility of the carrier connection.

Higher-dimensional unification in the Kaluza–Klein program [69, 72] sought to generate gauge fields and matter modes from the geometry of compact spatial extra dimensions, but encounters compactification instabilities and fails to generate chiral four-dimensional fermions without ad-hoc insertions. From another direction, condensed matter and analog gravity programs, exemplified by Volovik’s quantum liquid models [122] and Levin and Wen’s string-net condensates [82], demonstrate that emergent gauge bosons and chiral fermions can arise as low-energy quasiparticle excitations near Fermi points or topological ground states. These condensed-matter analogues rely on an antecedent non-relativistic Hamiltonian of microscopic atoms or spins residing in an external laboratory space and laboratory time, leaving the origin of the continuum and universal Lorentz covariance unaddressed.

In contrast, the present formulation derives physical matter directly from the constrained variational geometry of the carrier bundle $TM \oplus N$ over the base manifold, without extra dimensions, without antecedent matter Lagrangians, and without external laboratory backgrounds. Standard additive actions $S = S_g + S_m$ represent effective phenomenological decouplings that hold when metric geometry and matter sources are approximated as independent variables. In the full variational dynamics, unconstrained geometry contracts toward finite-time curvature singularity collapse. Preserving a non-degenerate physical continuum requires the configuration space to satisfy unilateral domain constraints $g(\Psi) \leq b$. Physical matter is the localized reactive constraint strain tensor $\lambda_{\mu\nu}^* \geq 0$ generated at this saturation threshold to halt metric compression, identifying $T_{\mu\nu}^m \equiv \lambda_{\mu\nu}^*$ directly through d’Alembert’s principle and Signorini variational contact equilibrium. Across singular Whitney defect strata $\partial\Sigma_k$ where the carrier response operator drops rank, connection curvature jump discontinuities trap chiral spin-1/2 Dirac zero modes via the Jackiw–Rebbi mechanism [64] and the Callias index theorem [17]. This establishes the microscopic-to-macroscopic operator identity $T_{\mu\nu}^m(x) \equiv \langle \Omega_0 | : \hat{T}_{\mu\nu}^D(x) : | \Omega_0 \rangle \equiv \lambda_{\mu\nu}^*(x)$, generating stable, localized fermionic matter from the intrinsic incompressibility of carrier geometry. At the universal matter spacing floor $b_3 \approx 0.60$ fm, calibrated empirically to the non-perturbative nuclear scale, localized transition currents concentrate across an invariant surface skin depth. This provides the exact geometric foundation for the physical non-factorization of leptonic wave functions discovered by Cadeddu et al. [16], wherein interaction-site wave-function attenuation resolves capture anomalies directly without postulating unobserved sterile particle sectors.

2.7. Curvature-Forced Mass Generation. In the Standard Model of particle physics, mass is introduced through the Brout–Englert–Higgs mechanism [36, 59]. This mechanism postulates an extrinsic complex scalar doublet carrying an ad-hoc tachyonic potential $V(\phi) = \lambda(\phi^\dagger\phi - v^2/2)^2$, which triggers spontaneous symmetry breaking and assigns masses through Yukawa couplings. In pure non-Abelian gauge theory, proving the existence of a non-perturbative mass gap $\Delta > 0$ in four-dimensional Euclidean space without an extrinsic scalar field constitutes the Clay Millennium Problem [67]. In non-perturbative gauge quantization, Gribov established that eliminating gauge-orbit redundancies requires restricting configuration space to a fundamental domain bounded by the Gribov horizon [49].

Riemannian spectral geometry demonstrates that spatial geometry can enforce uniform spectral floors. On complete negatively curved Riemannian manifolds, McKean’s spectral theorem proves that the Laplacian spectrum satisfies the strict

lower bound $\lambda_0 \geq |K|/4$ [91], while Cheeger's isoperimetric inequality bounds the first eigenvalue by geometric bottlenecks [24]. While the singularity theorems of Penrose and Hawking [56, 55] show that attractive convergence inevitably produces geodesic incompleteness in unconstrained spacetime, the present construction couples matter and gauge fields through constrained variational equilibrium. In general relativity, Witten established the positivity of gravitational energy by evaluating the Lichnerowicz–Weitzenböck curvature identity along harmonic spinors [132]. Non-Abelian gauge energy couples quadratically to gravitation and forces strict negative normal curvature $K = -b_4^2 < 0$. This negative curvature activates McKean's Riemannian spectral theorem [91], establishing the McKean curvature spectral floor,

$$(9) \quad \lambda_1(-\Delta_h + V) \geq \frac{|K|}{4} = \frac{b_4^2}{4} > 0.$$

which guarantees an exact non-perturbative physical mass gap $m = |b_4|/2 > 0$ without invoking an extrinsic Higgs field. Apparent zero-mode degeneracies on flat three-tori are eliminated by Hodge–de Rham cohomology [60, 30, 124] on physical S^3 spatial slices, while the Karush–Kuhn–Tucker matter spacing constraint [70, 76] breaks dilatation invariance, lifting the radial mode into the positive spectrum and preventing geodesic collapse.

3. THE CONFIGURATION SPACE OF METRIC GERMS

The configuration space of the field theory is constructed from equivalence classes of Riemannian metric germs on tubular neighborhoods of the zero section in a vector bundle, prior to the introduction of any connection, horizontal distribution, or spacetime embedding. Tracing the first jets of these metric germs recovers the gauge connection and the extrinsic shape tensor as exact geometric outputs of the ambient Levi-Civita connection. Factoring out the kernel of the Koszul morphism establishes an affine isomorphism between equivalence classes of metric germs and physical connection-shape pairs. Restricting the fiber metric to unimodular volume and equipping localized Sobolev variations with an integrated tensor inner product yields a complete, positive-definite infinite-dimensional Riemannian configuration manifold admitting an orthogonal decomposition into gauge and gravitational sectors.

3.1. Metric Germs on Tubular Neighborhoods of the Zero Section. Geometric configurations are defined on open neighborhoods of the zero section without presupposing a gauge connection or an ambient spacetime metric. Let $\pi: N \rightarrow M$ be a smooth real vector bundle of rank r over a smooth Riemannian base manifold (M, h) of dimension d , and let g_N denote a smooth fiber metric on N . Let $s: M \rightarrow N$ denote the canonical zero section. The differential ds and vertical differentiation define the canonical splitting of the ambient tangent bundle restricted to the zero section,

$$(10) \quad T_{s(x)}N = ds(T_x M) \oplus N_x, \quad x \in M.$$

The intersection of the two summands is trivial because the projection satisfies $d\pi \circ ds = \text{id}_{TM}$ while $d\pi$ annihilates vertical vectors. The restricted tangent bundle

$$(11) \quad E = s^*TN \cong TM \oplus N$$

inherits the block direct-sum metric

$$(12) \quad g_0 = h \oplus g_N.$$

This direct-sum decomposition is fixed canonically by the vector-bundle structure of N without selecting an auxiliary horizontal distribution.

The configuration space $\text{Met}_{g_0}(N, s)$ consists of germs of smooth symmetric positive-definite covariant 2-tensors on open neighborhoods of $s(M)$ in N that restrict to g_0 along the zero section. Two such tensors defined on open neighborhoods $U_1, U_2 \subset N$ of $s(M)$ define the same germ if they coincide on some open neighborhood $U_3 \subset U_1 \cap U_2$ containing $s(M)$. The restriction condition enforces that for all sections u, v of E , the pairing evaluates to $g_0(u, v)$ at every point of the zero section.

The space $\text{Met}_{g_0}(N, s)$ is non-empty. Let $\{(\phi_l, U_l)\}$ be a locally finite trivializing coordinate cover of M with subordinate smooth partition of unity $\{\psi_l\}$, and let y^a denote standard linear fiber coordinates on $N|_{U_l}$. On each local chart, the tensor

$$(13) \quad H_l = h_{ij}(x) dx^i dx^j + (g_N)_{ab}(x) dy^a dy^b$$

is smooth and positive-definite on the total chart domain $U_l \times \mathbb{R}^r$. The global sum

$$(14) \quad H_0 = \sum_l (\psi_l \circ \pi) H_l$$

defines a smooth, positive-definite metric on an open neighborhood of $s(M)$ satisfying $s^*H_0 = g_0$ on E . Consequently, the metric-first configuration space is well-defined and non-empty without presupposing a connection.

3.2. Levi-Civita Connections from First-Jet Bundles. Every metric germ in $\text{Met}_{g_0}(N, s)$ determines a unique ambient Levi-Civita connection whose tangential restriction to the zero section yields an assembled metric connection on E . Let $J_{g_0}^1$ denote the space of smooth first jets along $s(M)$ of symmetric covariant 2-tensors whose value is g_0 and whose tangential derivatives along $s(M)$ match the derivatives of g_0 . The space $J_{g_0}^1$ is an affine bundle modeled on the vector space of sections of

$$(15) \quad V = N^* \otimes \text{Sym}^2(E^*).$$

A difference between two metric germs with identical restriction g_0 has vanishing tangential derivatives along $s(M)$, so its first derivative along the zero section depends only on transverse normal directions.

For each metric germ $H \in \text{Met}_{g_0}(N, s)$, the ambient Levi-Civita connection ∇^H on the tubular neighborhood is uniquely determined by Koszul's formula [74]. For any vector field X tangent to M and any section Z of E , the covariant derivative

$$(16) \quad \mathcal{C}_{H,X}Z = \left(\nabla_{ds(X)}^H \tilde{Z} \right) \Big|_{s(M)}$$

is independent of the local extension $\tilde{Z}$ of Z off the zero section. This derivative defines a linear connection $\mathcal{C}_H$ on the vector bundle $E \rightarrow M$. Because ∇^H is metric-compatible with H , the assembled connection preserves the direct-sum metric g_0 , satisfying $X(g_0(Y, Z)) = g_0(\mathcal{C}_{H,X}Y, Z) + g_0(Y, \mathcal{C}_{H,X}Z)$.

Projecting the assembled connection onto the bundle summands TM and N determines its block structure. The tangent projection restricted to tangent vector fields has vanishing torsion, matching the unique Levi-Civita connection $D =$

$\text{LC}(h)$. The normal projection on normal sections defines a g_N -compatible connection A on N . The off-diagonal projection defines the extrinsic shape tensor

$$(17) \quad K(X, Y) = \text{pr}_N (\mathcal{C}_{H,X}(ds(Y))).$$

Torsion-freeness of ∇^H enforces symmetry in the tangent arguments, $K(X, Y) = K(Y, X)$, identifying K as a section of $\text{Sym}^2(T^*M) \otimes N$. Compatibility with g_0 requires the upper-right block to be the metric adjoint $-K^\dagger$. As a consequence, the assembled connection adopts the canonical matrix block decomposition

$$(18) \quad \mathcal{C}_H = \begin{pmatrix} D & -K^\dagger \\ K & A \end{pmatrix}.$$

The internal gauge connection A and the extrinsic shape tensor K arise as direct mathematical consequences of the metric first jet.

3.3. Quotient Identification via the Koszul Morphism. Physical field configurations of connection-shape pairs (A, K) correspond bijectively to equivalence classes of metric first jets modulo the kernel of the Koszul morphism. Let W denote the bundle of physical field variations,

$$(19) \quad W = (T^*M \otimes \mathfrak{so}(N, g_N)) \oplus (\text{Sym}^2(T^*M) \otimes N).$$

The variation of the metric first jet produces a linear bundle map $\sigma: V \rightarrow W$. For any section $D \in \Gamma(V)$, let $X, Y \in TM$ and $v, w \in N$. Evaluating the Koszul formula on the difference tensor D yields the components $\sigma(D) = (\alpha, k)$, where

$$(20) \quad \begin{aligned} g_N(k(X, Y), v) &= -\frac{1}{2} D(v; X, Y), \\ g_N(\alpha_X v, w) &= \frac{1}{2} [D(v; X, w) - D(w; X, v)]. \end{aligned}$$

The tensor α_X is manifestly skew-symmetric with respect to g_N , confirming that α takes values in $\mathfrak{so}(N, g_N)$.

The bundle morphism σ admits an explicit linear right inverse $R: W \rightarrow V$. For any pair $(\alpha, k) \in W$, define the tensor $R(\alpha, k) \in V$ by its evaluation on arguments $v \in N$ and $U_1, U_2 \in E$,

$$(21) \quad \begin{aligned} R(\alpha, k)(v; X, Y) &= -2 g_N(v, k(X, Y)), \\ R(\alpha, k)(v; X, w) &= g_N(\alpha_X v, w), \\ R(\alpha, k)(v; u, w) &= 0, \end{aligned}$$

extended by symmetry in the second and third arguments. Substituting these definitions into the morphism components (20) demonstrates that

$$(22) \quad \sigma \circ R = \text{id}_W.$$

The skew-symmetry of α ensures that $g_N(\alpha_X v, w) - g_N(\alpha_X w, v) = 2g_N(\alpha_X v, w)$, canceling the factor of one-half.

The kernel $\ker \sigma$ consists of all variations in V that produce zero first-order variation in the assembled connection. These correspond to transverse metric derivatives with vanishing tangent-tangent evaluation and cross evaluations that are symmetric in the two normal entries, while all-normal components remain arbitrary. Passing to the quotient space

$$(23) \quad Q_{g_0} = J_{g_0}^1 / \ker \sigma$$

establishes an affine isomorphism between equivalence classes of metric first jets and physical connection-shape configurations (A, K) .

3.4. Unimodular Fiber Normalization Bounds. Redundant local scale variations are removed by restricting the fiber metric to unimodular volume, isolating dynamical field degrees of freedom from unobservable scale rescalings. Let $\mathcal{U}_0$ denote the subspace of metric germs whose fiber metric g_N satisfies the unimodular constraint

$$(24) \quad \det(g_N) = 1$$

with respect to a fixed background volume element. For any traceless metric perturbation $u \in \Gamma(\text{Sym}^2(N^*))$ satisfying $\text{tr}(u) = 0$, the perturbed determinant expands in terms of the eigenvalues of u .

The determinant admits an exact quadratic remainder bound on bounded domains. Writing the perturbed fiber metric as $g_N(I + u)$, the determinant satisfies

$$(25) \quad \det(I + u) = \exp(\text{tr}(\log(I + u))) = 1 + \text{tr}(u) + r(u).$$

Because $\text{tr}(u) = 0$, the linear variation vanishes identically. On any compact window $U \subset M$ where the operator norm satisfies $\|u\| \leq 1/2$, expanding the logarithm into its Taylor series yields

$$(26) \quad |r(u)| \leq 2\|u\|^2.$$

This bound prevents finite-volume blowup and scale degeneracies across the configuration space.

The unimodular restriction is compatible with the Koszul quotient. The tangential and cross derivatives that define the morphism σ do not interfere with the algebraic constraint $\det(g_N) = 1$. The physical configuration space is therefore identified with the unimodular quotient manifold

$$(27) \quad \mathcal{Q} = \mathcal{U}_0 / \ker \sigma.$$

This quotient isolates the true dynamical degrees of freedom of the field.

3.5. Positive Configuration Metrics on Sobolev Domains. Variational calculus and dynamical response flows on the configuration space require a complete Hilbert manifold structure. Let $U \subset M$ be a relatively compact regular domain with smooth boundary. For an integer index $s > \dim(U)/2 + 2$, the Sobolev space $H^s(U, W)$ continuously embeds into $C^2(U, W)$ by the Sobolev embedding theorem, ensuring that second-order differential operators and curvature tensors remain continuous. Variational tangent vectors are restricted to the subspace $H_0^s(U, W)$ of sections vanishing on ∂U .

For tangent variations $(\delta A, \delta K) \in H_0^s(U, W)$, the configuration metric G is defined by the integrated inner product

$$(28) \quad G((\delta A, \delta K), (\delta A, \delta K)) = \int_U \left[\frac{1}{2} |\delta A|_{h \otimes g_N}^2 + |\delta K|_{h \otimes g_N}^2 \right] d\mu_h,$$

where $d\mu_h = \sqrt{\det h} d^d x$ is the Riemannian volume element of (M, h) , and the pointwise norms are induced by the Riemannian metrics h and g_N . The factor of one-half on the gauge connection variation reflects the standard normalization of Lie algebra generators.

The metric G is strictly positive-definite and coercive on $H_0^s(U, W)$. For any non-zero variation, the integral is strictly positive, vanishing if and only if $\delta A = 0$

and $\delta K = 0$ almost everywhere on U . The induced L^2 -type topology makes the tangent space a pre-Hilbert space, whose Sobolev completion yields the model Hilbert space for the infinite-dimensional configuration manifold. This positive-definiteness stands in contrast to the indefinite signature $(-, +, +, +, +, +)$ of the DeWitt supermetric in canonical geometrodynamics [31], providing kinematic stability and preventing negative kinetic energies.

3.6. Orthogonal Decomposition along the Distinguished Shape Line. The configuration tangent space splits orthogonally under the metric G into independent gauge connection and gravitational shape sectors. Let T_A and T_K denote the respective subspaces of W ,

$$(29) \quad \begin{aligned} T_A &= \Omega^1(M, \mathfrak{so}(N, g_N)), \\ T_K &= \Gamma(\text{Sym}^2(T^*M) \otimes N). \end{aligned}$$

At any reference configuration (A, K) , the cross-pairing between a pure gauge variation $(\delta A, 0)$ and a pure shape variation $(0, \delta K)$ evaluates to

$$(30) \quad G((\delta A, 0), (0, \delta K)) = 0.$$

The cross-term vanishes because δA and δK take values in orthogonal tensor representations over the base manifold. Consequently, the tangent space of the configuration manifold admits the orthogonal direct-sum decomposition

$$(31) \quad T_{(\mathcal{U}_0 / \ker \sigma)} = T_A \oplus^\perp T_K.$$

Within the shape sector T_K , geometric scaling defines a distinguished one-dimensional kinematic trajectory. Let $K_0 \in T_K$ be a fixed non-zero reference shape tensor. The one-parameter family

$$(32) \quad K(q) = q K_0, \quad q \in \mathbb{R}$$

traces the distinguished shape line in configuration space. Along this line, the extrinsic geometry scales uniformly with the scalar parameter q . This distinguished line provides the internal kinematic coordinate along which the autonomous curvature response flow generates macroscopic spacetime history, separating dynamical evolution from static background geometry.

4. GEOMETRIC ENERGY AND POLAR CALCULUS

The dynamics of field configurations on the unimodular quotient manifold $\mathcal{Q} = \mathcal{U}_0 / \ker \sigma$ is governed by a quadratic curvature-energy functional. Evaluating the curvature of the assembled connection on the restricted bundle $E = TM \oplus N$ establishes an exact three-sector Gauss–Codazzi–Ricci decomposition projecting the curvature onto tangent, mixed, and normal subbundles. The weak first variation of this functional yields the symmetrized Euler–Lagrange response fields driving autonomous configuration flow. The second variation generates a self-adjoint four-block Hessian operator whose off-diagonal blocks demonstrate that internal gauge connection perturbations and metric shape variations are dynamically coupled through the Codazzi sector. On constant-rank strata, the mixed response block admits an invariant polar decomposition that isolates positive radial amplitudes and determines a canonical complex structure generating internal phase symmetries, while singular derivative obstructions at rank-changing defect strata restrict the validity of the differential calculus to stratified domains.

4.1. The Curvature-Energy Functional on Direct-Sum Bundles. Field dynamics is formulated through a gauge-invariant variational functional measuring the total geometric curvature of the assembled connection. Let $E = TM \oplus N$ carry the direct-sum metric $g_0 = h \oplus g_N$, and let $\mathcal{C}$ be an assembled connection on E preserving g_0 . The curvature two-form $F_{\mathcal{C}} \in \Omega^2(M, \text{End}(E))$ defines the quadratic curvature-energy functional on any relatively compact window $U \subset M$ by the integrated Hilbert–Schmidt norm,

$$(33) \quad \Phi = \frac{1}{2} \int_U |F_{\mathcal{C}}|^2 d\mu_h.$$

The pointwise norm $|F_{\mathcal{C}}|^2 = \text{tr}(F_{\mathcal{C}}^* F_{\mathcal{C}})$ is computed using the Riemannian volume element $d\mu_h = \sqrt{\det h} d^d x$ and the metric g_0 on the fibers of E .

The functional Φ is invariant under fiber-preserving gauge transformations and bounded from below. For any smooth bundle automorphism $\mathcal{U} = \text{id}_{TM} \oplus u_N$ preserving the metric g_0 , the curvature transforms tensorially as $F_{\mathcal{U} \cdot \mathcal{C}} = \mathcal{U} F_{\mathcal{C}} \mathcal{U}^{-1}$. The cyclic property of the trace ensures that $|F_{\mathcal{U} \cdot \mathcal{C}}|^2 = |F_{\mathcal{C}}|^2$ at every point of M , establishing gauge invariance. Furthermore, the integrand is non-negative, ensuring that $\Phi \geq 0$, with $\Phi = 0$ if and only if the assembled connection is flat. This functional provides the variational principle for field responses without postulating extrinsic action densities.

4.2. Three-Sector Gauss–Codazzi–Ricci Curvature Decomposition. The curvature two-form decomposes algebraically into three geometric sectors governing tangent, mixed, and normal bundle geometries. Let $X, Y \in \Gamma(TM)$ be smooth tangent vector fields on M . The curvature two-form is evaluated by the commutator

$$(34) \quad F_{\mathcal{C}}(X, Y) = [\mathcal{C}_X, \mathcal{C}_Y] - \mathcal{C}_{[X, Y]}.$$

Substituting the canonical block decomposition of the assembled connection $\mathcal{C}_X$ from equation (18) into this commutator yields the exact matrix block representation

$$(35) \quad F_{\mathcal{C}}(X, Y) = \begin{pmatrix} T_{XY} & -M_{XY}^\dagger \\ M_{XY} & N_{XY} \end{pmatrix}.$$

Direct algebraic multiplication identifies each sector with its geometric counterpart in subbundle geometry. The tangential Gauss sector takes the form

$$(36) \quad T_{XY} = R_{XY}^h - K_X^\dagger K_Y + K_Y^\dagger K_X,$$

where $R_{XY}^h = [D_X, D_Y] - D_{[X, Y]}$ is the intrinsic Riemann curvature tensor of (M, h) . This sector combines intrinsic curvature with quadratic contractions of the shape tensor, recovering the classical Gauss equation for submanifolds. The mixed Codazzi sector is given by

$$(37) \quad M_{XY} = (\nabla_X K)_Y - (\nabla_Y K)_X,$$

where the covariant derivative ∇ acts on tangent indices with D and on normal indices with A , via $(\nabla_X K)_Y = D_X^A(K_Y) - K_{D_X Y}$. This tensor measures the covariant bending of the shape field under the coupled connection, matching the Codazzi–Mainardi equations. The normal Ricci sector adopts the expression

$$(38) \quad N_{XY} = F_{XY}^A - K_X K_Y^\dagger + K_Y K_X^\dagger,$$

where $F_{XY}^A = [A_X, A_Y] - A_{[X, Y]}$ is the field strength two-form of the internal gauge connection A , modified by normal commutators of the shape operator.

Decomposing the curvature-energy functional (33) across these three sectors gives the explicit integral form

$$(39) \quad \Phi = \frac{1}{2} \int_U (|T|^2 + 2|M|^2 + |N|^2) d\mu_h,$$

where $|M|^2 = \text{tr}(M^\dagger M)$ accounts for both off-diagonal corners. The functional thus balances intrinsic tangential geometry, mixed covariant bending, and normal gauge curvature.

4.3. Weak First Variation for Symmetrized Euler–Lagrange Response.

Computing the gradient of Φ on the Sobolev Hilbert space $H_0^s(U, W)$ determines the Euler–Lagrange response fields driving configuration evolution. Let $v = (a, k) \in H_0^s(U, W)$ denote field perturbations, with $a \in \Omega^1(M, \mathfrak{so}(N))$ an internal connection variation and $k \in \Gamma(\text{Sym}^2(T^*M) \otimes N)$ a shape variation. The variation of the assembled connection defines the block endomorphism

$$(40) \quad c_v = \begin{pmatrix} 0 & -k^\dagger \\ k & a \end{pmatrix}.$$

The linearized curvature operator $L_v = D_C c_v$ generates the first-order variation of F_C .

The weak first variation of the functional Φ is evaluated through the integrated inner product

$$(41) \quad D\Phi[v] = \langle F_C, L_v \rangle = \langle D_C^\dagger F_C, c_v \rangle.$$

Boundary integrals vanish identically because variations in $H_0^s(U, W)$ have compact support in U . Setting $Z = D_C^\dagger F_C$, the raw variational derivatives are obtained by projection onto the respective bundle components. The raw connection variation evaluates to $E_A = Z_N$, while the raw shape variation evaluates to $E_0(X, Y) = \text{pr}_N Z_X(Y, 0)$.

Physical shape perturbations belong to the symmetric tensor space $\text{Sym}^2(T^*M) \otimes N$. The symmetrized Euler–Lagrange shape response is obtained by applying the orthogonal symmetrization projector P_+ , yielding

$$(42) \quad Y_A = E_A, \quad Y_K = 2 P_+ E_0.$$

The factor of two accounts for the dual pairing against symmetric test tensors. The response vector (Y_A, Y_K) represents the negative gradient of the curvature energy under the configuration metric G .

4.4. Four-Block Weak Hessians in Coupled Gauge-Gravitational Sectors.

The second weak variation of Φ defines a self-adjoint Hessian operator whose off-diagonal blocks couple connection and shape fluctuations. Let $v = (a, k)$ and $w = (b, l)$ be variations in $H_0^s(U, W)$. The second variation is given by the bilinear form

$$(43) \quad \mathcal{H}(v, w) = \langle L_v, L_w \rangle + \langle F_C, [c_v \wedge c_w] \rangle.$$

Decomposing this form into pure gauge and shape components yields the complete four-block Hessian operator

$$(44) \quad \mathcal{H} = \begin{pmatrix} \mathcal{H}_{AA} & \mathcal{H}_{AK} \\ \mathcal{H}_{KA} & \mathcal{H}_{KK} \end{pmatrix}.$$

The diagonal blocks govern uncoupled gauge and shape fluctuations. The connection block evaluates to

$$(45) \quad \mathcal{H}_{AA}(a, b) = 2\langle L_a^M, L_b^M \rangle + \langle D_A a, D_A b \rangle + \langle N, [a \wedge b] \rangle,$$

where $L_a^M(X, Y) = a_X K_Y - a_Y K_X$ and D_A is the covariant exterior derivative on N -valued forms. The shape block evaluates to

$$(46) \quad \mathcal{H}_{KK}(k, l) = \langle L_k^T, L_l^T \rangle + 2\langle d_\nabla k, d_\nabla l \rangle + \langle L_k^N, L_l^N \rangle + \langle T, R_{kl}^T \rangle + \langle N, R_{kl}^N \rangle,$$

where L_k^T and L_k^N are algebraic shape contractions and R_{kl} denotes commutator contractions of test fields.

The off-diagonal cross blocks are self-adjoint and satisfy

$$(47) \quad \mathcal{H}_{KA}(k, b) = \mathcal{H}_{AK}(b, k) = 2\langle d_\nabla k, L_b^M \rangle + \langle L_k^N, D_A b \rangle + 2\langle M, R_{kb}^M \rangle.$$

This cross-coupling is non-vanishing. On a local coordinate patch with flat connection background and constant shape tensor $K_1 = qf_1$ with $q \neq 0$, evaluating $\mathcal{H}_{AK}(b, k)$ against linear test variations yields an exact evaluation of $2q$. Because $q \neq 0$, the cross-coupling block $\mathcal{H}_{AK}$ does not vanish. Consequently, gauge fields and shape distortions cannot be decoupled dynamically, as the mixed Codazzi curvature generates irreducible cross-interactions.

4.5. Polar Decomposition and the Canonical Complex Structure. On any fiber where the shape operator is non-zero, the mixed response block admits an invariant polar decomposition isolating radial amplitudes and a canonical complex structure. Fix a point $p \in M$ and a non-zero tangent vector $X \in T_p M$. The mixed response operator on $T_p M \oplus N_p$ is defined by the block matrix

$$(48) \quad B_X = \begin{pmatrix} 0 & -K_X^\dagger \\ K_X & 0 \end{pmatrix}.$$

The operator B_X is skew-adjoint, satisfying $B_X^\dagger = -B_X$. Its square is negative semidefinite, producing the block-diagonal positive semidefinite operator

$$(49) \quad -B_X^2 = \begin{pmatrix} K_X^\dagger K_X & 0 \\ 0 & K_X K_X^\dagger \end{pmatrix} \geq 0.$$

By the finite-dimensional spectral theorem for self-adjoint operators, there exists a unique non-negative square root

$$(50) \quad S_X = \sqrt{-B_X^2} = \begin{pmatrix} (K_X^\dagger K_X)^{1/2} & 0 \\ 0 & (K_X K_X^\dagger)^{1/2} \end{pmatrix} \geq 0.$$

The operator S_X defines the radial amplitude tensor. The kernel $\ker B_X = \ker K_X \oplus \ker K_X^\dagger$ identifies uncoupled directions. The canonical carrier support is the orthogonal complement

$$(51) \quad \mathcal{H} = (\ker B_X)^\perp = \text{im}(K_X^\dagger) \oplus \text{im}(K_X).$$

On the subspace $\mathcal{H}$, the amplitude operator S_X is strictly positive and invertible.

Let $S_X^\dagger$ denote the Moore–Penrose pseudoinverse [92, 95] of S_X , satisfying $S_X^\dagger S_X = P_{\mathcal{H}}$, where $P_{\mathcal{H}}$ is the orthogonal projection onto $\mathcal{H}$. The canonical phase operator is defined by

$$(52) \quad J_X = B_X S_X^\dagger = S_X^\dagger B_X.$$

Restricted to the active carrier $\mathcal{H}$, this operator satisfies the defining properties of an almost complex structure,

$$(53) \quad \boxed{J_X^2 = -\mathbf{I}_{\mathcal{H}}, \quad J_X^\dagger = -J_X, \quad J_X^\dagger J_X = \mathbf{I}_{\mathcal{H}}.}$$

The operator J_X acts as an orthogonal isometry on $\mathcal{H}$, and the mixed block factors as $B_X = S_X J_X = J_X S_X$. In each singular vector two-plane $\text{span}(t, n)$ spanned by a tangent singular vector t and its normal image $n = K_X t / \|K_X t\|$, the exponential $\exp(\theta J_X)$ parameterizes the compact rotation group $SO(2) \cong U(1)$. The polar decomposition thus extracts internal phase rotations directly from the geometry of the mixed curvature block.

4.6. Stratified Regularity and Obstructions across Rank-Changing Defect Strata. The differential calculus of the polar factors S_X and J_X is smooth on constant-rank strata but obstructed across rank-changing defect strata. Let $U_q \subset U$ be an open stratum where $\text{rank}(K_X) = q$ is locally constant. On U_q , the active carrier $\mathcal{H}$ forms a smooth vector subbundle of $TM \oplus N$, and the non-zero eigenvalues of $-B_X^2$ are separated from zero by a uniform spectral gap $\delta > 0$. The orthogonal projector onto $\mathcal{H}$ is given by the Cauchy resolvent contour integral

$$(54) \quad P_{\mathcal{H}} = \frac{1}{2\pi i} \oint_{\gamma} (zI + B_X^2)^{-1} dz,$$

where $\gamma \subset \mathbb{C} \setminus \{0\}$ is a positively oriented smooth contour enclosing the non-zero spectrum. Standard functional calculus guarantees that S_X , $S_X^\dagger$, and J_X are smooth bundle endomorphisms on U_q .

Across defect strata where $\text{rank}(K_X)$ changes, differentiability fails. Consider the one-parameter crossing family $K(x) = x$ over $\mathbb{R}$. The squared operator $-B^2(x) = x^2 \mathbf{I}_2$ produces the radial amplitude $S(x) = |x| \mathbf{I}_2$. The one-sided derivative of $S(x)$ at $x = 0$ evaluates to $-\mathbf{I}_2$ from the left and $+\mathbf{I}_2$ from the right, generating a finite jump discontinuity

$$(55) \quad \Delta S' = S'_+(0) - S'_-(0) = 2\mathbf{I}_2.$$

Simultaneously, the phase operator $J(x) = \text{sgn}(x) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has no continuous extension to $x = 0$.

This singular obstruction establishes the stratified topology of the configuration domain. The global configuration space is rigorously structured as a Whitney stratified space [130, 129, 47] partitioned into smooth manifolds of constant rank. Consequently, all resolvent expansions, spectral derivatives, and variation calculi are strictly restricted to constant-rank strata. At singular defect strata where the rank changes, field trajectories transition between distinct smooth strata, preserving mathematical consistency across the complete configuration space.

5. CURVATURE AND AUTONOMOUS RESPONSE EVOLUTION

Dynamical evolution of field configurations is governed by an autonomous first-order response flow on the unimodular quotient manifold $\mathcal{Q} = \mathcal{U}_0 / \ker \sigma$, formulated prior to the introduction of any ambient spacetime manifold atlas, coordinate clocks, or Lorentzian signature. Driven by the symmetrized Euler–Lagrange response fields of the curvature-energy functional Φ , the response vector field generates unique,

continuously differentiable solution trajectories on Sobolev domains. The monotonic dissipation of curvature energy defines a dissipative parameter τ establishing a thermodynamic and dynamical arrow of time along the trajectory, while preserving local jet coherence. Unilateral variational contact equilibrium (governed by the Karush–Kuhn–Tucker theorem for convex cones [70, 76]) concurrently generates a reactive constraint strain trajectory that exhibits gauge invariance under lineality shifts and preserves the strict positive-definiteness of the configuration metric. Lifting this flow to cotangent phase space represents the response dynamics as the Poisson derivation of a signed Legendre Hamiltonian, providing the classical geometric foundation for quantum operator constraints.

5.1. Dynamics on the Configuration Quotient. Field dynamics is defined on the configuration quotient space without presupposing a pre-existing spacetime manifold atlas or external coordinate clocks. Classical formulations define temporal evolution by selecting a time coordinate within a four-dimensional Lorentzian manifold $(M, g_{\mu\nu})$ and evolving Cauchy initial data through foliation vector fields. Because the existence of a global smooth spacetime manifold atlas remains an open conditional outcome, obstructed by topological branchings and causal singularities, postulating an ambient spacetime manifold to define field evolution introduces a circular dependency.

The dynamical trajectory $t \mapsto \Psi(t) = (A(t), K(t))$ is therefore constructed entirely within the fiber-bundle configuration quotient $\mathcal{Q}$. The parameter t designates an intrinsic evolution coordinate parameterizing gradient flow curves on configuration space. Macroscopic four-dimensional spacetime and its extrinsic geometry are reconstructed as the historical foliation of this generated trajectory along the distinguished shape line, eliminating circular dependence on an assumed manifold atlas.

5.2. Autonomous Response Evolution Generators on Sobolev Domains.

The curvature-energy gradient defines an autonomous vector field on Sobolev spaces that generates unique continuously differentiable configuration histories. On the Sobolev Hilbert space $H_0^s(U, W)$ with integer index $s > \dim(U)/2 + 2$, field configurations $\Psi = (A, K)$ generate the Euler–Lagrange response fields (Y_A, Y_K) derived from the assembled curvature two-form in equation (42). The autonomous response generator $\mathcal{V}(\Psi) = (\mathcal{V}_A, \mathcal{V}_K)$ projects these fields onto the subspace preserving unilateral domain constraints and gauge invariance,

$$(56) \quad \mathcal{V}_A = -Y_A + \mathcal{P}_{\ker D_A} Y_A, \quad \mathcal{V}_K = -Y_K + \mathcal{P}_{\ker d_\nabla} Y_K.$$

where $\mathcal{P}_{\ker D_A}$ and $\mathcal{P}_{\ker d_\nabla}$ denote orthogonal projectors onto the respective differential kernels under the configuration metric G .

Configuration trajectories satisfy the autonomous first-order dynamical system

$$(57) \quad \boxed{\frac{d\Psi}{dt} = \mathcal{V}(\Psi(t)), \quad \Psi(0) = \Psi_0 \in \mathcal{Q}.}$$

The generator $\mathcal{V}$ is locally Lipschitz continuous from $H^s(U, W)$ to $H^{s-2}(U, W)$ due to the Sobolev multiplication algebra property for $s > \dim(U)/2 + 2$. By the Cauchy–Lipschitz and Picard–Lindelöf theorems on Banach spaces [97, 84, 77], for every initial configuration $\Psi_0 \in \mathcal{Q}$, there exists a maximal interval of existence $[0, T_{\max})$ and a unique continuously differentiable trajectory $t \mapsto \Psi(t)$. Configuration histories are thereby generated dynamically by internal geometric forces.

5.3. Admissible Cones, Dissipative Flow, and the Arrow of Time. The autonomous response flow is strictly dissipative off equilibrium, generating an intrinsic, monotonically increasing parameter that defines the arrow of time. Along any solution trajectory $\Psi(t)$, the time derivative of the curvature-energy functional Φ evaluates to

$$(58) \quad \frac{d\Phi}{dt} = D\Phi[\mathcal{V}] = \langle Y, \mathcal{V} \rangle = -\|Y - \mathcal{P}Y\|_G^2 \leq 0.$$

Because the configuration metric G is positive-definite, the time derivative of Φ is strictly non-positive, vanishing if and only if the projected response fields vanish identically.

The dissipated curvature energy defines a monotone dissipation parameter,

$$(59) \quad \tau(t) = -\Phi(\Psi(t)) \implies \frac{d\tau}{dt} = \|Y - \mathcal{P}Y\|_G^2 \geq 0.$$

The parameter $\tau(t)$ is monotonically increasing along every non-equilibrium trajectory, providing an internal arrow of time that arises directly from geometric dissipation without external clocks. Furthermore, the flow preserves the jet coherence condition

$$(60) \quad c(t) = \Gamma(t) - \frac{\partial_\phi \varepsilon(t)}{\varepsilon(t)} = 0.$$

where $\Gamma(t)$ is the Christoffel trace and $\varepsilon(t)$ is the local spacing scale. Differentiating $c(t)$ along the trajectory demonstrates that variational flow preserves compatibility between connection coordinates and spacing variations.

5.4. Concurrent Reactive Strain Trajectories at Constrained Stationarity. Constrained variational stationarity along the configuration trajectory concurrently generates a unique path of reactive constraint strain tensors enforcing geometric constraints. At each point of the trajectory $\Psi(t)$, unilateral contact equilibrium [70, 76] determines the concurrent reactive strain path

$$(61) \quad t \mapsto (\nu(t), \lambda(t), \mu(t)), \quad \lambda(t) = \lambda_1(t) + i\lambda_2(t) \in \mathbb{C}.$$

The irreversibility parameter $\nu(t) \geq 0$ acts as a reactive normal strain governing irreversible dissipation, resisting compression whenever the trajectory saturates the threshold of the admissible variation cone.

The complex strain component $\lambda(t) = \lambda_1(t) + i\lambda_2(t)$ constrains the extremal faces of the cone. The real and imaginary components are assembled into a complex variable through the canonical complex structure J_X certified in the polar decomposition, establishing internal phase rotation of the reactive strain components. The scalar multiplier $\mu(t) \in \mathbb{R}$ enforces the equality constraint $c(t) = 0$. These components determine the reactive strain variables governing variational stability along the trajectory.

5.5. Reactive Strain Lineality Invariance under Metric Positivity Preservation. The response dynamics is invariant under a one-parameter group of gauge lineality shifts and preserves the strict positive-definiteness of the configuration metric. The reactive strain tensor admits the lineality shift

$$(62) \quad \lambda_1(t) \mapsto \lambda_1(t) - a, \quad \Delta\nu(t) \mapsto \Delta\nu(t) + a, \quad a \in \mathbb{R}.$$

for any real constant a . Because the configuration equations of motion depend exclusively on the invariant combination $\lambda_1(t) + \Delta\nu(t)$, the physical solution trajectory $\Psi(t)$ is invariant under the strain lineality group $(\mathbb{R}, +)$.

The configuration metric remains strictly positive-definite along the flow. Let $G(t) = h(t) \oplus g_N$ denote the metric on $\mathcal{Q}$. Because the initial metric $G(0)$ is strictly positive-definite with minimal eigenvalue bounded away from zero on the compact domain U , and because the solution path $\Psi(t)$ is continuous on $[0, T_{\max})$, the minimal eigenvalue of $G(t)$ remains strictly positive on every compact subinterval $[0, T_1] \subset [0, T_{\max})$. This stability rules out premature metric collapse and maintains the Riemannian structure of configuration space throughout the evolution.

5.6. Cotangent Phase Space for Signed Hamiltonian Derivations. Lifting the response flow to cotangent phase space represents field evolution as a Poisson derivation generated by a signed Legendre Hamiltonian. The cotangent bundle $T^*\mathcal{Q}$ carries canonical coordinates $(\Psi, \pi) = (A, K, \pi_A, \pi_K)$ and the symplectic two-form $\omega = \int_U [\delta\pi_A \wedge \delta A + \delta\pi_K \wedge \delta K] d\mu_h$. The cotangent kinetic energy is defined by the dual metric pairing

$$(63) \quad T(\Psi, \pi) = \frac{1}{2} \langle \pi, G^\sharp \pi \rangle.$$

where $G^\sharp$ denotes the inverse configuration metric on momenta.

The Legendre transform of the configuration Lagrangian selects the signed Hamiltonian function

$$(64) \quad H_R = T - \Phi.$$

The negative sign before the curvature energy Φ reflects the hyperbolic signature of gravitational response constraints. The associated Hamiltonian vector field X_{H_R} defines the autonomous Poisson derivation on smooth functionals,

$$(65) \quad \delta_{H_R}(F) = \{H_R, F\} = X_{H_R}(F).$$

This classical derivation satisfies the derivation property $\delta_{H_R}(F_1 F_2) = \delta_{H_R}(F_1) F_2 + F_1 \delta_{H_R}(F_2)$. It provides the exact classical generator that induces the canonical commutation relation algebra and the positive Hamiltonian constraint operator.

6. SPACETIME RECONSTRUCTION AND GRAVITATIONAL CONSTRAINTS

The classical conception of spacetime as an ambient container is replaced by a chronological foliation. Spatial geometry represents the relational equilibrium of the coupled shape and gauge degrees of freedom, while temporal duration is the observed parameter of autonomous gradient dissipation. Prior to this reconstruction, dynamics takes place exclusively on the infinite-dimensional configuration quotient space $\mathcal{Q}$. No ambient manifold atlas, no background Lorentzian metric, and no pre-existing spacetime coordinates are presupposed.

6.1. Spacetime Foliation from Autonomous Trajectories. The autonomous Picard–Lindelöf trajectory [97, 84, 77] generates macroscopic spacetime as its chronological trace, realizing the globally hyperbolic causal structure established by Hawking and Ellis [55]. By the local well-posedness established in Section 5, the autonomous gradient flow determines a unique maximal solution curve $t \mapsto \Psi(t) =$

$(h(t), K(t), A(t))$ on the configuration quotient $\mathcal{Q}$ over the interval of existence $I = (\tau_-, \tau_+)$. The spacetime history manifold is the topological product leaf

$$(66) \quad M = \Sigma \times I, \quad h(t) = \pi^* h(\tau)|_{\tau=t} \in \text{Sym}_{>0}^2(T^*\Sigma).$$

equipped with the canonical projection $\pi: M \rightarrow \Sigma$ and temporal coordinate $t = \tau$.

At each temporal slice $\Sigma_t = \Sigma \times \{t\}$, the induced spatial Riemannian metric is the configuration germ $h(t)$. Spacetime is assembled strictly as the continuous 1-parameter family of spatial metric germs foliated along the autonomous flow parameter. The manifold M possesses no independent physical existence apart from the historical sequence of spatial metric configurations.

6.2. The Distinguished Shape Line ($q = 1$) and Identical Ricci Flatness.

A one-dimensional normal shape subbundle identically eliminates normal curvature obstructions and guarantees symmetric extrinsic curvature. From the minimal internal rank classification $r = 3$ with complex carrier plane dimension $p = 1$, the real normal shape line $N\Sigma$ has rank $q = 1$. The normal bundle is spanned by a unit section ν with norm $\langle \nu, \nu \rangle = \sigma_N = \pm 1$.

Because $N\Sigma$ is one-dimensional, its structure group reduces to the discrete identity group $SO(1) = \{1\}$. The normal connection 1-form $\omega^\perp$ vanishes identically, forcing the normal curvature 2-form and the shape operator commutator to vanish,

$$(67) \quad \omega^\perp \equiv 0, \quad R^\perp(X, Y)\nu \equiv 0, \quad [A_\nu, A_\nu] \equiv 0 \quad \forall X, Y \in T\Sigma.$$

for all tangent vector fields $X, Y \in T\Sigma$. Consequently, the Ricci equation of submanifold geometry is satisfied as a trivial identity, leaving no algebraic obstructions on the extrinsic shape tensor.

Metric compatibility and the torsion-free property of the ambient Levi-Civita connection ensure that the shape operator A_ν is self-adjoint with respect to the spatial metric h . This self-adjointness defines the symmetric extrinsic curvature tensor

$$(68) \quad \Pi(X, Y) = h(A_\nu X, Y) = \Pi(Y, X).$$

Every geometric requirement for embedding a Riemannian hypersurface with extrinsic curvature Π is satisfied without extrinsic fine-tuning.

6.3. Contravariant Metric Forms and Admissible Causal Cone Matching.

The causal light cone of Lorentzian spacetime is identical to the extremal threshold of the cone of admissible configuration variations. The tangent bundle TM carries two canonical bilinear forms,

$$(69) \quad g = N^2 dt^2 + \varepsilon^2 h_{ij} dx^i dx^j, \quad \eta = N^2 dt^2 - \varepsilon^2 h_{ij} dx^i dx^j.$$

where g is the positive-definite Riemannian observation metric evaluating total operational length, and η is the effective metric with physical Lorentzian signature $(+, -, -, -)$. Here N denotes the lapse function, and ε scales the spatial metric.

Evaluating the quadratic form η on an arbitrary tangent displacement vector $V = \frac{dt}{d\lambda} \partial_t + \frac{dx^i}{d\lambda} \partial_i$ matches the unilateral constraint clearance $S(V)$ [70, 76],

$$(70) \quad \boxed{\eta(V, V) = (\Delta\phi)^2 - \varepsilon^2 \|\Delta L\|_h^2, \quad S(V) = \Delta\phi - \varepsilon \|\Delta L\|_h.}$$

where $\Delta\phi = Ndt$ represents the phase progression and $\|\Delta L\|_h = \sqrt{h_{ij} dx^i dx^j}$ represents spatial displacement.

This identity establishes an exact correspondence between relativistic causal structure and variational admissibility across three exhaustive regimes.

- (1) Strictly admissible internal transitions with $S(V) > 0$ correspond to time-like displacement vectors with $\eta(V, V) > 0$.
- (2) Saturated admissible cone thresholds with clearance $S(V) = 0$ correspond to null rays with $\eta(V, V) = 0$.
- (3) Inadmissible transitions with $S(V) < 0$ correspond to spacelike separations with $\eta(V, V) < 0$.

Relativistic causality is therefore not an independent postulate of geometry. It is the physical manifestation of dissipative gradient-flow irreversibility on configuration space.

6.4. Gauss–Codazzi Integrability and the ADM Constraint Identification. Projecting the ambient curvature tensor along tangent and normal directions generates the classical gravitational initial-data constraints and isolates four evolution degrees of freedom. Let $\vec{n} = \frac{1}{N}(\partial_t - N^i \partial_i)$ denote the future-directed unit normal vector field to the leaves Σ_t , where N^i is the shift vector. The extrinsic curvature matches the temporal velocity of the configuration metric via the Arnowitt–Deser–Misner relation

$$(71) \quad K_{ij} = \frac{1}{2N}(\partial_t h_{ij} - \nabla_i N_j - \nabla_j N_i) = \Pi_{ij}.$$

Projecting the ambient Riemann curvature tensor of (M, η) onto tangent and normal components yields the Gauss and Codazzi compatibility equations. Tracing these projections with the inverse spatial metric h^{ij} yields the gravitational Hamiltonian constraint

$$(72) \quad \boxed{\mathcal{H}_g = R_\Sigma - K_{ij}K^{ij} + K^2 = 16\pi G\rho.}$$

and the spatial momentum constraint

$$(73) \quad \mathcal{M}_i = \nabla_j K_i^j - \nabla_i K = 8\pi G J_i.$$

where $K = h^{ij}K_{ij}$ is the trace of extrinsic curvature, and ρ and J_i denote the energy and momentum densities of matter sources.

By the fundamental theorem of hypersurfaces due to Bonnet, Tenenblat, and Hartman–Wintner [12, 54, 116], the satisfaction of the Gauss, Codazzi, and Ricci equations guarantees the local existence of an isometric immersion of (Σ, h) with extrinsic curvature K into (M, η) . On a three-dimensional spatial slice, the local metric-germ initial data possesses 8 kinematic degrees of freedom, consisting of 6 spatial metric components and 2 unimodular normal directions. Subtracting the 4 initial-data constraints in (72) and (73) leaves exactly four physical evolution degrees of freedom for the gravitational-shape sector.

6.5. The Jacobi Operator and Deterministic Metric Reconstruction. Metric perturbations are reconstructed deterministically from extrinsic curvature variations through the boundedly invertible Jacobi observation operator on transverse-traceless carrier support. Linearizing the extrinsic curvature response $\delta\Pi$ with respect to metric variations δh defines the self-adjoint Jacobi observation operator [65]

$$(74) \quad \mathcal{J}(\delta h) = -\frac{1}{2}\Delta_L \delta h + \frac{1}{2}\nabla\nabla(\text{Tr } \delta h) + \frac{1}{2}R * \delta h + \frac{1}{4}\Pi_0(\Pi_0 * \delta h).$$

where Δ_L is the Lichnerowicz Laplacian [83], $\nabla\nabla(\text{Tr } \delta h)$ represents the longitudinal divergence, $R*\delta h$ denotes the algebraic Riemann curvature coupling, and Π_0 is the background extrinsic curvature.

Pure gauge deformations generated by spatial diffeomorphisms lie in the kernel $\ker B_X = \text{im}(\mathcal{L}) = \ker \sigma$. By the Berger–Ebin and York orthogonal decomposition [10, 136], the physical variation space decomposes into gauge and transverse-traceless carrier components,

$$(75) \quad \text{Sym}^2(T^*\Sigma) = \ker B_X \oplus \mathcal{H}, \quad \mathcal{H} = (\ker B_X)^\perp = \ker(\text{div}) \cap \ker(\text{Tr}).$$

where the carrier subspace $\mathcal{H} = (\ker B_X)^\perp = \ker(\text{div}) \cap \ker(\text{Tr})$ consists of transverse-traceless symmetric tensors.

On the carrier support $\mathcal{H}$, the Lichnerowicz Laplacian is strictly elliptic with a strictly positive discrete spectrum bounded below by $\mu_{\min} > 0$. By the Lax–Milgram theorem [78], the restricted operator $\mathcal{J}|_{\mathcal{H}}$ possesses a bounded inverse,

$$(76) \quad \delta h = \mathcal{J}^{-1}(\delta \Pi), \quad \|\mathcal{J}^{-1}\|_{\mathcal{B}(\mathcal{H})} \leq \frac{1}{\mu_{\min}}.$$

This bounded inversion achieves deterministic metric recovery $\delta h = \mathcal{J}^{-1}(\delta \Pi)$. An observer measuring extrinsic curvature fluctuations recovers the underlying metric variations uniquely without statistical path integration or Gaussian ensembles.

6.6. Stratified Spacetime Regularity Across Rank-Changing Interfaces.

Spacetime is locally smooth on regular strata and globally structured as a Whitney stratified pseudo-Riemannian manifold [130, 129, 47]. On any open stratum $\Sigma_k \subset \Sigma$ where the response operator B_X maintains locally constant rank k , the non-zero singular values are bounded away from zero by a uniform spectral gap. Holomorphic functional calculus guarantees that the polar differentials (dS_X, dJ_X) and the Jacobi inverse $\mathcal{J}^{-1}$ are C^∞ -smooth, ensuring that the local spacetime leaf $M_k = \Sigma_k \times I$ is a smooth Lorentzian manifold.

Across rank-changing singular defect strata $\partial\Sigma_k$, the singular values cross zero. At these defect strata, the derivative of the amplitude tensor exhibits a derivative discontinuity in the singular values and the complex structure incurs a discrete phase jump,

$$(77) \quad \Delta S' = 2, \quad \Delta J = 2J_0.$$

where J_0 denotes the canonical symplectic phase generator. Because the derivatives of the metric and connection exhibit jump discontinuities across $\partial\Sigma_k$, the spacetime Christoffel symbols fail C^1 regularity at the interfaces.

This structure characterizes the global manifold structure of reconstructed spacetime. A smooth manifold atlas is realized locally on regular strata $M_{\text{reg}} = \bigcup_k M_k$, but is obstructed globally as a single smooth C^∞ manifold atlas. The physical spacetime M is rigorously a Whitney stratified pseudo-Riemannian manifold [130, 129, 47] whose smooth Lorentzian leaves are interconnected across singular defect strata $\partial\Sigma_k$.

6.7. Comparative Positioning Among Emergent Spacetime Approaches.

Reconstructed bulk spacetime is deterministic, local, and bulk-direct, systematically distinguished from boundary dualities, polymer discreteness, and coarse-graining approximations across five major programs.

In holographic duality and quantum error-correcting tensor networks [88, 87, 121, 102, 110, 14, 94, 53, 35, 112], bulk spacetime is dual to boundary entanglement entropy, requiring an asymptotic negative cosmological constant, an auxiliary conformal boundary, and quantum error-correcting codes. In the present setting, macroscopic (3+1)-dimensional Lorentzian spacetime is generated directly in the bulk without an auxiliary boundary, without negative cosmological constant assumptions, and without non-local boundary entanglement dualities. Spacetime is the chronological foliation of the Picard–Lindelöf trajectory on the metric-germ quotient space, where lapse, shift, and extrinsic curvature are direct realizations of the autonomous configuration gradient flow along the distinguished shape line $q = 1$. Relativistic causality is not decoded from boundary entanglement entropy or quantum error correction codes, but from the admissible cone threshold of the constrained variational contact dynamics [70, 76].

In the space-from-Hilbert-space program [20, 19], spatial geometry and metric perturbations are reconstructed from quantum mutual information matrices and modular Hamiltonians on an abstract Hilbert space. The present construction does not attempt to extract spatial Riemannian geometry from an ungrounded abstract Hilbert space. The spatial Riemannian metric germ h is the initial equilibrium configuration, while temporal evolution, Lorentzian signature, and the ADM constraints are generated deterministically by the autonomous response flow without statistical entanglement matrices or modular Hamiltonian approximations.

In canonical loop quantum gravity [101, 100], continuous metrics are replaced by $SU(2)$ holonomies and flux variables on non-separable polymer Hilbert spaces, yielding discrete area and volume spectra while confronting persistent Hamiltonian non-closure anomalies. The present construction derives continuous 4D Lorentzian geometry and the ADM constraints from smooth metric germs and the Koszul differential right inverse on standard separable functional domains, bypassing polymer discreteness and non-closure anomalies.

In the amplituhedron program [2], local spacetime and unitarity are eliminated in favor of positive Grassmannian polytope volumes. The present theory does not discard local spacetime or unitarity. Instead, local spacetime is derived as the exact historical foliation of the configuration trajectory, preserving causal light cones and local gauge invariance while maintaining unitary operator representations.

In the quantum renormalization group approach [80, 81], an emergent holographic direction is generated from energy-scale coarse graining in matrix models. The evolution parameter τ in the present formulation is not an energy-coarse-graining scale of an auxiliary boundary matrix field, but the autonomous parameter of the curvature gradient flow on the metric-germ quotient space, preserving exact microscopic geometric data along the entire trajectory without coarse-graining cutoffs or large- N matrix approximations.

7. VARIATIONAL MATTER COUPLINGS AND INTERFACE GEOMETRY

Physical matter is the localized reactive constraint strain of saturated geometric configuration constraints on the configuration quotient space. Matter fields are not introduced as an extrinsic phenomenological addition; matter arises through variational constraint mechanics from the geometric curvature functional.

7.1. Matter as Reactive Constraint Strain of Saturated Geometry. The stress-energy-momentum tensor arises via constrained variational mechanics as the

optimal reactive constraint strain tensor enforcing geometric configuration admissibility. Under autonomous gradient flow, geometry contracts toward curvature collapse with volume shrinkage rate $\partial_t \ln \sqrt{\det h} \leq -c_0 \|K\|_h^2$. Unilateral configuration constraints ($\det h \geq v_0 > 0$, $\varepsilon \geq b_3 > 0$) halt this geometric collapse, generating an outward normal reactive constraint strain tensor $\lambda_{\mu\nu}^* \geq 0$ along saturated interfaces. In constrained variational mechanics on the configuration quotient $\mathcal{Q}$, the saddle-point formulation takes the form

$$(78) \quad \inf_{\Psi \in \mathcal{Q}} \Phi(\Psi) + \sup_{\lambda \in \Lambda^*} \langle \lambda, \mathcal{C}(\Psi) \rangle.$$

where Λ^* is the cone of positive reactive constraint strain tensors enforcing unilateral domain constraints.

By Signorini contact mechanics and d'Alembert's principle, the reactive constraint strain tensor $\lambda_{\mu\nu}^*$ evaluates the reaction force opposing geometric compression at saturated boundaries. Variation of the effective action with respect to the Lorentzian metric η determines the symmetric Hilbert stress-energy-momentum tensor

$$(79) \quad T_{\mu\nu} = \frac{2}{\sqrt{-\det \eta}} \frac{\delta S_m}{\delta \eta^{\mu\nu}} = \lambda_{\mu\nu}^*.$$

Topologically, connection curvature jumps $\Delta J_{\mu\nu}^a$ across singular Whitney defect strata $\partial \Sigma_k$ act as topological domain walls, trapping chiral spin-1/2 Dirac zero modes via the Jackiw–Rebbi and Callias index theorems [64, 17]. The physical matter stress-energy tensor is identically the expectation value of the trapped Dirac field, exactly matching the reactive constraint strain tensor,

$$(80) \quad \begin{aligned} T_{\mu\nu}^m(x) &\equiv \langle \Omega_0 | : \hat{T}_{\mu\nu}^D(x) : | \Omega_0 \rangle \equiv \lambda_{\mu\nu}^*(x), \\ \text{supp}(T^m) &\equiv \text{supp}(\lambda^*) \equiv \{x \in \Sigma \mid \varepsilon(x) = b_3\}. \end{aligned}$$

Matter is therefore not an ad hoc substance placed upon spacetime. It is the localized reactive constraint strain tensor that enforces geometric incompressibility under the autonomous response flow.

7.2. The Quadratic Gravitation Theorem. The effective gravitational coupling is strictly negative and quadratic in the fourth response coefficient, enforcing universal gravitational attraction. In the expansion of the effective energy functional about the unimodular equilibrium, the second-order variation with respect to conformal scale yields the gravitational action term

$$(81) \quad \boxed{S_g[\eta] = -b_4^2 \int_M R[\eta] d\mu_\eta.}$$

where b_4 is the fourth response amplitude, and $R[\eta]$ is the spacetime scalar curvature.

The negative sign in (81) reflects the hyperbolic signature of the gravitational kinetic sector in the DeWitt supermetric on $\mathcal{Q}$. Comparing this variation with the standard Einstein–Hilbert action determines the Newton gravitational constant

$$(82) \quad \boxed{G = \frac{1}{16\pi b_4^2} > 0.}$$

Because $b_4 \neq 0$ on the physical configuration space, G is strictly positive, ensuring that the induced gravitational interaction is universally attractive.

7.3. Coupled Cauchy Data and Energy Conditions. Coupled gravitational and matter initial data satisfy the weak, dominant, and strong energy conditions on every spatial slice. The complete initial-data package on the spatial manifold Σ is specified by the septuple

$$(83) \quad (\Sigma, h, K, \rho, J_i, A, E).$$

where h is the spatial Riemannian metric, K is the extrinsic curvature tensor, ρ is the energy density, J_i is the momentum current density, and (A, E) are the gauge connection and electric field sections on the internal carrier bundle.

Because the underlying configuration metric G is strictly positive-definite on the quotient space $\mathcal{Q}$, the reactive stress-energy tensor satisfies

$$(84) \quad T_{\mu\nu} u^\mu u^\nu \geq 0, \quad -T^\mu{}_\nu u^\nu \in \bar{V}^+ \quad \forall u \in V^+.$$

for every future-directed timelike vector $u \in V^+$, where $\bar{V}^+$ denotes the closed future causal cone. Metric positivity on configuration space guarantees energy positivity of matter sources without auxiliary energy postulates.

7.4. Einstein–Maxwell Constraints via GCR Integrability. Gauss–Codazzi–Ricci integrability of the coupled bundle $E = TM \oplus N$ automatically generates the complete Einstein–Maxwell constraint equations. Expanding the ambient curvature tensor of the assembled connection $\nabla = (D, -K^\dagger; K, A)$ on the direct sum bundle yields the Maxwell stress-energy tensor $T_{\mu\nu}^{\text{EM}} = F_{\mu\alpha} F_\nu{}^\alpha - \frac{1}{4} \eta_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$.

Projecting the curvature identities onto tangent and normal directions yields the coupled gravitational Hamiltonian constraint

$$(85) \quad R_\Sigma - K_{ij} K^{ij} + K^2 = 16\pi G(\rho_g + \rho_{\text{EM}}).$$

and the coupled spatial momentum constraint

$$(86) \quad \nabla_j K_i^j - \nabla_i K = 8\pi G(J_i^g + J_i^{\text{EM}}), \quad D_i E^i = \rho_e.$$

The divergence of the momentum constraint vanishes identically on the constraint surface modulo the internal Gauss law $D_i E^i = \rho_e$, confirming the exact integrability of the coupled system.

7.5. Reflected Yang–Mills Equations and Local Cauchy Development.

The autonomous response flow on the gauge sector generates the spacetime Yang–Mills equations and guarantees local Cauchy development. Projecting the Hamiltonian response derivation δ_{H_R} onto the gauge components (A, π_A) yields the second-order field equations

$$(87) \quad D^\mu F_{\mu\nu} = J_\nu^m.$$

where D^μ is the spacetime covariant gauge derivative, and J_ν^m is the conserved matter current generated by internal phase variations.

By the Choquet–Bruhat and Geroch theorem for coupled hyperbolic systems [42, 27, 26], initial data satisfying constraints (85), (86), and the Gauss law admit a unique maximal Cauchy development

$$(88) \quad (M, \eta, A) = \text{Dev}(\Sigma, h, K, A, E).$$

The coupled spacetime manifold (M, η) together with the gauge field A is smooth and globally hyperbolic.

7.6. Conformal Symmetry Breaking and the Mass Floor. The unilateral matter spacing floor $b_3 > 0$ breaks conformal scale invariance and eliminates unconstrained dilatations. Under conformal rescalings of the spatial metric $h \mapsto e^{2\omega}h$, the Maxwell curvature energy $\int \|F\|^2 d\mu_h$ remains invariant, while the extrinsic curvature trace coupling associated with b_3 transforms non-trivially,

$$(89) \quad \delta_\omega \left[b_3 \int_\Sigma (\text{Tr } K)^2 d\mu_h \right] = b_3 \int_\Sigma [3(\text{Tr } K)^2 + 6(\text{Tr } K)\vec{n}(\omega)] e^{3\omega} d\mu_h \neq 0.$$

where $\vec{n}(\omega)$ is the normal derivative of the conformal factor along the flow.

This non-vanishing variation breaks global scale symmetry. Coupled with the quadratic curvature response b_4 , it generates an explicit spectral floor for the associated differential operators on spatial sections,

$$(90) \quad \lambda_1(-\Delta_h + V) \geq \frac{b_4^2}{4} = \frac{b_4^2}{4} > 0.$$

The existence of this strictly positive spectral floor $\lambda_1 > 0$ provides the geometric basis for the McKean spectral gap [91] and particle mass generation developed in Section 11.

8. GENERATED GAUGE LAW AND STABILIZER SELECTION

Internal gauge symmetry groups are selected as the exact stabilizers of geometric structures on the carrier bundle. Internal Lie groups are not postulated by heuristic symmetry choices; the gauge group is uniquely determined as the maximal compact subgroup preserving metric positivity, the polar complex structure, and interface stability on the configuration quotient space.

8.1. Minimal Carrier Rank and Subbundle Splitting. The internal normal bundle N admits an invariant real rank classification $r = 3$ splitting into an active complex transverse plane and an invariant normal shape line. Over the three-dimensional spatial manifold Σ , the total Serre–Swan carrier bundle [106, 111] is the direct sum $E = TM \oplus N$. To support the four-block weak Hessian, non-trivial curvature energy, the polar decomposition $B_X = S_X J_X$, and the distinguished normal shape line $q = 1$ simultaneously, the minimal total real rank of N is uniquely $r = 3$.

This minimal rank requirement establishes the canonical orthogonal Whitney sum decomposition

$$(91) \quad N = N_{\mathbb{C}} \oplus N_{\mathbb{R}}, \quad \text{rank}_{\mathbb{R}} N_{\mathbb{C}} = 2, \quad \text{rank}_{\mathbb{R}} N_{\mathbb{R}} = 1.$$

where $N_{\mathbb{C}}$ has real rank $2p = 2$ and carries the canonical almost-complex structure J_X , while $N_{\mathbb{R}}$ has real rank $q = 1$ and spans the normal shape line along the unit section ν .

The fiber metric g_N on the internal carrier bundle decomposes into invariant block diagonal form,

$$(92) \quad g_N = g_{N_{\mathbb{C}}} \oplus \sigma_N, \quad \sigma_N = \pm 1.$$

where $g_{N_{\mathbb{C}}}$ is a positive-definite metric on the complex carrier plane, and $\sigma_N = \pm 1$ characterizes the signature along the normal line.

8.2. The Transverse Plane Stabilizer $U(1)$ and Electromagnetic Selection.

The automorphism group preserving the polar complex structure on the carrier plane is uniquely the compact abelian Lie group $U(1)$. The canonical almost-complex structure J_X satisfies $J_X^2 = -I$ on the two-dimensional real subbundle $N_{\mathbb{C}} \cong \mathbb{R}^2$. Any linear isometry $g \in SO(2)$ commuting with J_X belongs to the centralizer

$$(93) \quad gJ_X = J_Xg \implies g \in C_{SO(2)}(J_X) \cong U(1).$$

Variations along this centralizer preserve the polar decomposition of the response operator and maintain the positive-definiteness of the configuration metric G . The associated connection 1-form takes values in the Lie algebra $\mathfrak{u}(1)$,

$$(94) \quad A = A_\mu dx^\mu \otimes Q, \quad Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

where Q is the standard skew-symmetric generator representing the imaginary unit. The curvature 2-form $F = dA$ coincides with the electromagnetic field tensor whose sourced field equations were established in Section 7.

8.3. The Weak Isospin Stabilizer $SU(2)$ from Orthogonal Slices. Rotating the active complex plane within the three-dimensional normal fiber N_x selects the special unitary group $SU(2)$ as the non-abelian isospin stabilizer. The internal fiber N_x is a three-dimensional Euclidean space whose isometry group is $SO(3)$. Lifting this orthogonal action to the associated spin bundle yields the universal double covering group $\text{Spin}(3) \cong SU(2)$.

The infinitesimal generators $\tau_a = \frac{1}{2}\sigma_a$ satisfy the Lie algebra commutation relations

$$(95) \quad [\tau_a, \tau_b] = i\epsilon_{abc}\tau_c, \quad \tau_a = \frac{1}{2}\sigma_a \quad (a, b, c \in \{1, 2, 3\}).$$

where ϵ_{abc} is the Levi-Civita permutation symbol. These generators preserve the Riemannian fiber metric,

$$(96) \quad g_N(\tau_a v, w) + g_N(v, \tau_a w) = 0 \quad \forall v, w \in N_x.$$

for all vectors $v, w \in N_x$. The group $SU(2)$ is the unique maximal connected non-abelian stabilizer of three-dimensional fiber geometry, identifying it with the weak isospin gauge symmetry.

8.4. Maximal Color Stabilizer $SU(3)$ under Complex Stratification. Complex threefold stratification of the tangent-carrier bundle forces the special unitary color group $SU(3)$ as the maximal stable symmetry. Across regular geometric strata Σ_k , the complete tangent-carrier geometry admits a complexification spanned by three independent complex planes $N_{\mathbb{C}}^{\oplus 3} \cong \mathbb{C}^3$.

The maximal group of orientation-preserving linear automorphisms preserving the standard Hermitian metric on $\mathbb{C}^3$ with unit determinant is $SU(3)$. Its infinitesimal generators satisfy the Gell-Mann Lie algebra

$$(97) \quad [\lambda_A, \lambda_B] = 2if_{ABC}\lambda_C \quad (A, B, C \in \{1, \dots, 8\}).$$

where f_{ABC} are the completely antisymmetric structure constants of $\mathfrak{su}(3)$. Every generator preserves the Hermitian metric,

$$(98) \quad h_{\mathbb{C}}(\lambda_A v, w) + h_{\mathbb{C}}(v, \lambda_A w) = 0 \quad \forall v, w \in \mathbb{C}^3.$$

for all vectors $v, w \in \mathbb{C}^3$. Any extension to higher-rank unitary groups violates the minimal carrier rank bound or introduces unstable negative directions in the configuration Hessian.

8.5. The Gauge Group Classification Theorem and Geometric Selection.

The exact product group $G = [\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)]/\Gamma$ is the unique maximal compact Lie group realized by stable automorphisms of the metric-germ configuration space. Combining the individual stabilizer classifications across the carrier subbundles yields the total gauge group

$$(99) \quad G = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}, \quad \Gamma \cong \mathbb{Z}_6.$$

where $\Gamma \cong \mathbb{Z}_6$ is the discrete central subgroup generated by the common center elements of the three factors.

The assembled gauge connection decomposes uniquely along these stabilizer subalgebras,

$$(100) \quad A = A_3^A \lambda_A + A_2^a \tau_a + A_1 Q.$$

This decomposition demonstrates that the Standard Model gauge group is not an empirical postulate inserted into an action functional. It is the unique maximal geometric stabilizer that preserves metric positivity, polar complex structures, and Gauss–Codazzi–Ricci integrability on the carrier bundle.

8.6. Obstruction of Exceptional and Higher-Rank Groups. Exceptional Lie groups and higher-rank orthogonal or unitary groups are obstructed by positive-definiteness and finite carrier rank. Hypothetical gauge extensions such as $SU(5)$, $SO(10)$, or E_6 confront two structural obstructions.

First, realizing representations of $SU(5)$ or $SO(10)$ requires carrier bundle fibers of real dimension at least 10. The minimal carrier rank theorem forces

$$(101) \quad r = 3 < 10 \implies \mathfrak{g} \not\supset \mathfrak{su}(5), \mathfrak{so}(10), \mathfrak{e}_6.$$

Any fiber expansion beyond $r = 3$ introduces negative eigenvalues into the configuration metric G , violating the prerequisite for a Riemannian configuration space.

Second, evaluating the weak Hessian on generators outside the stabilizer algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ reveals hyperbolic instabilities,

$$(102) \quad \mathrm{spec} \left(\mathcal{H}|_{\mathfrak{g} \setminus (\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1))} \right) \cap (-\infty, 0) \neq \emptyset.$$

These negative directions cause the Picard–Lindelöf autonomous flow [97, 84, 77] to experience finite-time gradient blowup. Consequently, the observed gauge symmetries mark the exact mathematical threshold of stable geometric response.

9. SOURCE CHARACTERIZATION AND LOCAL CONSERVED CURRENTS

Conserved physical matter currents and localized charges are derived as surface Noether currents of geometric gauge stabilizers acting on the carrier bundle. Physical sources are not modeled as extrinsic continuous fluids, but arise from the geometric support of polar rank interfaces across stratified spacetime leaves.

9.1. Noether Currents of Geometric Gauge Stabilizers. Conserved matter currents are derived as the Noether currents associated with the infinitesimal automorphisms of the gauge stabilizer group. Under an infinitesimal gauge transformation $\delta_\xi A = D\xi$ parameterized by a section $\xi = \xi^a T^a \in \Gamma(\text{ad } P)$, the variation of the curvature energy functional Φ on the configuration quotient $\mathcal{Q}$ satisfies the divergence relation

$$(103) \quad \delta_\xi \Phi = \int_M \nabla_\mu J_a^\mu \xi^a d\mu.$$

where J_a^μ is the local Noether current density on the spacetime leaf M .

Evaluating this variation along the polar decomposition $B_X = S_X J_X$ and the assembled connection blocks (D, K, A) yields the explicit algebraic expression

$$(104) \quad J_\mu^a = \text{Tr}_N (\nabla_\mu J_X [T^a, J_X]) + \langle S_X T^a, K_\mu \rangle.$$

The conserved current is constructed directly from intrinsic geometric data on the carrier bundle without introducing external matter variables.

9.2. Local Charge Conservation and the Bianchi Identity. Covariant conservation of gauge currents is guaranteed identically by the contracted second Bianchi identity on the assembled carrier bundle. Computing the covariant divergence of the spacetime Yang–Mills equations $D^\mu F_{\mu\nu}^a = J_\nu^a$ yields

$$(105) \quad D^\mu J_\mu^a = 0.$$

This identity follows from the antisymmetry of the field strength tensor $F_{\mu\nu}$ combined with the Ricci identity for the gauge covariant derivative,

$$(106) \quad D^\mu D^\nu F_{\nu\mu}^a = \frac{1}{2} [F^{\mu\nu}, F_{\nu\mu}]^a \equiv 0.$$

The commutator term vanishes by the Jacobi identity for the Lie algebra structure constants. Consequently, local electric, isospin, and color charge currents are strictly conserved without imposing independent phenomenological continuity equations.

9.3. Charge Quantization from Polar Phase Periodicity. Physical charge quantization is mathematically forced by the compact periodicity of the polar complex structure on the carrier plane. The canonical almost-complex structure J_X generates a one-parameter isometry group whose exponential map satisfies the spectral periodicity condition

$$(107) \quad \exp(2\pi J_X) = I_{\mathcal{H}}.$$

on the transverse carrier subspace $\mathcal{H}$.

Because physical state sections must remain single-valued under this 2π rotation on the configuration quotient $\mathcal{Q}$, the holonomy of the connection around non-contractible loops is restricted to integer multiples of the 2π circle period. This topological constraint forces electric charge to satisfy the exact quantization condition

$$(108) \quad q \in q_0 \mathbb{Z}, \quad q_0 = \sqrt{\frac{\hbar c}{4\pi}}.$$

Continuous charge deformation is algebraically obstructed by the global topology of the carrier quotient space.

9.4. Current Localization on Rank-Crossing Interfaces. Physical matter sources are localized on singular defect strata where the response operator changes rank. On any regular stratum Σ_k , the response operator maintains constant rank k , and the field equations are source-free. Across the rank-changing defect strata $\partial\Sigma_k$, the singular values cross zero, creating jump discontinuities in the tangential gauge curvature,

$$(109) \quad [F_{\mu\nu}^a]_-^+ = \Delta J_{\mu\nu}^a \neq 0.$$

where $\Delta J_{\mu\nu}^a$ is the generated interface phase jump.

Applying the distributional divergence theorem across the singular defect stratum $\partial\Sigma_k$ yields a delta-function surface current density,

$$(110) \quad J_\Sigma^a = \delta(\partial\Sigma_k) \mathbf{n}^\mu [F_{\mu\nu}^a]_-^+.$$

where $\mathbf{n}^\mu$ is the unit normal vector to the interface. Localized point-like and string-like matter sources correspond to the topological defect strata of the underlying Whitney stratification.

9.5. Complete Stress-Energy-Current Conservation. The total divergence of the coupled gravitational and gauge stress-energy tensor vanishes identically modulo the matter current equations of motion. Decomposing the total stress-energy-momentum tensor into geometric, electromagnetic, and interface matter components yields

$$(111) \quad T_{\mu\nu} = T_{\mu\nu}^g + T_{\mu\nu}^{\text{EM}} + T_{\mu\nu}^m.$$

Evaluating the spacetime covariant divergence of $T_{\mu\nu}$ with respect to the Levi-Civita connection of (M, η) yields the conservation identity

$$(112) \quad \nabla^\mu T_{\mu\nu} = 0.$$

The Lorentz force term $F_{\nu\lambda}^a J_a^\lambda$ exerted by the gauge field on the localized current is balanced by the divergence of the geometric stress tensor. The coupled physical system exhibits exact four-momentum conservation across all regular strata and singular interfaces.

9.6. The Classical Sourced Cauchy Problem. Sourced Einstein–Yang–Mills initial data on stratified spaces admit a unique local-in-time Cauchy development. The complete initial-data system on the spatial slice Σ consists of the coupled Hamiltonian and momentum constraints together with the internal Gauss law,

$$(113) \quad \mathcal{H}_{\text{tot}} = 0, \quad \mathcal{M}_{\text{tot}} = 0, \quad D_i E^i = \rho_e.$$

On each regular stratum Σ_k , specifying initial data $(h, K, A, E) \in H^s(\Sigma_k) \times H^{s-1}(\Sigma_k)$ with $s > 5/2$ satisfying (113) guarantees the existence of a unique maximal Cauchy development

$$(114) \quad (M, \eta, A) \in C^0(I, H^s(\Sigma_k)) \cap C^1(I, H^{s-1}(\Sigma_k)).$$

matching across singular defect strata $\partial\Sigma_k$ via jump conditions (109) and (110). This establishes local well-posedness for the coupled field equations on stratified spacetime leaves.

10. OPERATIONAL RECOVERY AND OBSERVABLE RECONSTRUCTION

Geometric configurations and internal gauge fields are reconstructed operationally through measurement protocols executed by localized physical observers. Idealized coordinate charts are not assumed; metric and connection data are determined through proper time intervals, geodesic deviations, holonomy interferometry, and the bounded inversion of extrinsic curvature responses.

10.1. Operational Observer Frames and Proper Time. Physical observation is defined by an orthonormal tetrad carried along an observer timelike worldline in the reconstructed spacetime leaf. An observer is represented by a timelike worldline $\gamma(\tau) \subset M$ parameterized by proper time τ , equipped with an orthonormal frame $e_a^\mu(\tau)$ satisfying

$$(115) \quad \eta(e_a, e_b) = \eta_{ab} = \text{diag}(1, -1, -1, -1).$$

where $\eta_{ab} = \text{diag}(1, -1, -1, -1)$ is the Minkowski metric of the local rest frame.

The timelike vector $e_0 = u = \dot{\gamma}$ represents the observer four-velocity. The spatial triad e_i ($i = 1, 2, 3$) is propagated along the worldline according to the Fermi–Walker transport equation [38, 123]

$$(116) \quad \frac{D_F e_i}{d\tau} = \nabla_u e_i - \eta(a, e_i)u + \eta(u, e_i)a = 0.$$

where $a = \nabla_u u$ is the four-acceleration. The proper time elapsed along γ directly evaluates the effective Lorentzian metric η , providing the physical standard for temporal duration.

10.2. Radar Metric Reconstruction and Geodesic Deviation. Spatial Riemannian metric components are measured operationally through round-trip radar timing and geodesic deviation of test particles. An observer emits a null signal at proper time $\tau^{(1)}$, reflected at an event p , and received at proper time $\tau^{(2)}$. The operational radar distance is given by

$$(117) \quad d_R(p, \gamma) = \frac{1}{2}c(\tau^{(2)} - \tau^{(1)}).$$

where c is the invariant speed of light defined by the admissible cone threshold.

In a local Fermi normal coordinate neighborhood of γ , the spatial components of η expand as $g_{ij}(x) = \delta_{ij} - \frac{1}{3}R_{ikjl}(\tau)x^k x^l + O(|x|^3)$. Measuring the relative acceleration of nearby test particles with separation vector Y^μ evaluates the spacetime curvature tensor through the Jacobi geodesic deviation equation

$$(118) \quad \frac{D^2 Y^\mu}{d\tau^2} = R^\mu{}_{\nu\alpha\beta} u^\nu u^\alpha Y^\beta.$$

This operational protocol extracts the electric components of the Riemann curvature tensor without coordinate ambiguity.

10.3. Gauge Holonomy Interferometry and Wilson Loops. Internal gauge connections are reconstructed through phase shifts measured by Wilson loop interferometry around closed spatial contours. For any closed spatial loop $\gamma \subset \Sigma$, charged matter test particles transported around γ acquire a path-ordered holonomy

$$(119) \quad W_R(\gamma) = \text{Tr}_R \mathcal{P} \exp \left(i \oint_\gamma A \right).$$

where R denotes the representation of the gauge stabilizer group G , and $\mathcal{P}$ indicates path ordering.

For contractible loops, shrinking the contour area extracts the localized field strength tensor $F_{\mu\nu}^a$ in representation R ,

$$(120) \quad F_{\mu\nu}^a = \lim_{\text{Area}(\gamma) \rightarrow 0} \frac{\text{Tr}(T^a W_R(\gamma))}{\text{Area}(\gamma)}.$$

In the abelian sector, this reproduces the Aharonov–Bohm effect, while in the non-abelian sectors it reconstructs color and weak field strengths directly from phase-shift fringe patterns.

10.4. Deterministic Inversion of Extrinsic Curvature Observations. Observed extrinsic curvature perturbations deterministically reconstruct spatial metric germs through the regularized Jacobi inverse. An observer network monitoring extrinsic curvature variations δII via tidal tensor accelerations reconstructs the underlying metric germ perturbation through the linear equation

$$(121) \quad \delta h = \mathcal{J}^{-1}(\delta\text{II}).$$

On the transverse-traceless carrier subspace $\mathcal{H} = (\ker B_X)^\perp$, the self-adjoint operator $\mathcal{J}$ has a strictly positive discrete spectrum bounded by $[\mu_{\min}, \mu_{\max}]$. The operational condition number is strictly bounded,

$$(122) \quad \kappa(\mathcal{J}|_{\mathcal{H}}) = \frac{\mu_+}{\mu_-} < \infty.$$

This finite bound ensures that small measurement uncertainties in extrinsic curvature do not induce unbounded errors in the reconstructed spatial metric germ, establishing stable numerical and physical invertibility.

10.5. Measurement Stability Across Stratified Interfaces. Operational measurements remain stable across singular rank-changing interfaces through distributional matching conditions. While metric derivatives exhibit the singular-value derivative discontinuity $\Delta S' = 2$ and connection fields incur the phase jump $\Delta J = 2J_0$ across singular interfaces $\partial\Sigma_k$, proper time remains continuous and rectifiable,

$$(123) \quad \tau = \int_0^T \sqrt{\eta(\dot{\gamma}, \dot{\gamma})} dt \in C^0([0, T]).$$

Physical clock readings experience no discontinuous jumps when traversing singular interfaces. Similarly, operational radar distances between events across singular interfaces satisfy the uniform Lipschitz stability bound

$$(124) \quad |d_R(p_1) - d_R(p_2)| \leq C \|x_1 - x_2\|_h.$$

where $C > 0$ depends only on the metric bounds on regular strata. Physical measurements remain well-defined and stable across all topological defect strata.

10.6. Universal Measurement Resolution and Finite Carrier Cutoff. The minimal internal bundle rank and positive configuration metric establish an intrinsic lower bound on operational measurement resolution. Evaluating the canonical commutation relation on the cotangent bundle $T^*\mathcal{Q}$ reveals an operational uncertainty relation between metric perturbations and conjugate momenta,

$$(125) \quad \Delta h_{ij} \Delta \pi^{ij} \geq \frac{\hbar}{2}.$$

Attempting to resolve spatial locations with precision finer than the minimal geometric resolution scale $\ell_{\min}$ induces an extrinsic curvature back-reaction that exits the regular quotient $\mathcal{Q}$. This back-reaction enforces the universal lower bound on spatial measurement resolution

$$(126) \quad \Delta x \geq \ell_0 = \sqrt{\frac{G\hbar}{c^3}} > 0.$$

Physical space cannot be resolved into infinitesimal mathematical points by any operational measurement apparatus. This intrinsic geometric cutoff eliminates ultraviolet divergences without requiring ad hoc regulators.

11. CONCRETE PHYSICAL REALIZATIONS AND THE CURVATURE MASS GAP

Coupled geometric configurations generate explicit non-perturbative particle solutions and establish a strictly positive mass gap for non-abelian gauge fields. Solitons, colored black holes, and the Yang–Mills mass gap arise directly from the curvature energy functional and the conformal-breaking response floor without relying on lattice approximations or phenomenological cutoffs.

11.1. Spherically Symmetric Einstein–Yang–Mills Solitons. Spherically symmetric configurations of the coupled geometry generate the explicit Bartnik–McKinnon particle solitons [7] without horizons or singularities. For an $SU(2)$ gauge stabilizer acting on a static, spherically symmetric spacetime leaf, the metric and connection take the ansatz form

$$(127) \quad ds^2 = \mu(r)\sigma(r)^2 dt^2 - \frac{dr^2}{\mu(r)} - r^2 d\Omega^2, \quad A = (w(r) - 1)(\tau_1 d\theta + \tau_2 \sin\theta d\phi).$$

where $\mu(r)$ and $\sigma(r)$ parameterize the spacetime geometry, and $w(r)$ represents the non-abelian magnetic potential.

Substituting this ansatz into the coupled field equations established in Section 7 yields the autonomous radial system

$$(128) \quad \begin{aligned} r\mu' + \mu &= 1 - \frac{8\pi G}{g^2 r^2}(1 - w^2)^2 - \frac{16\pi G}{g^2}\mu(w')^2, \\ r^2\mu w'' + \left(r^2\mu' + \frac{r^2\mu\sigma'}{\sigma}\right)w' + w(1 - w^2) &= 0. \end{aligned}$$

subject to the regularity conditions $\mu(0) = 1$, $\sigma(0) = 1$, $w(0) = 1$ and asymptotic flatness $\mu(\infty) = 1$, $\sigma(\infty) = 1$, $w(\infty) = \pm 1$. This system admits a discrete sequence of smooth, globally regular, finite-mass soliton solutions, realizing particle-like field bundles from pure geometry.

11.2. Colored Black Holes and Non-Abelian Hair. Imposing outer event horizon boundary conditions selects stable colored black hole solutions with non-abelian gauge hair. Setting an event horizon at radius $r_h > 0$ requires the boundary conditions

$$(129) \quad \mu(r_h) = 0, \quad \mu'(r_h) > 0, \quad |w(r_h)| < 1.$$

with $\sigma(r_h)$ non-zero and finite.

Integrating the radial equations outward from r_h to spatial infinity proves the existence of static, asymptotically flat colored black holes. The total Arnowitt–Deser–Misner mass evaluates to

$$(130) \quad M_{\text{ADM}} = \frac{1}{2G} \int_{r_h}^{\infty} \left[\frac{(1-w^2)^2}{r^2} + 2\mu(w')^2 \right] \sigma(r) dr.$$

These solutions circumvent scalar no-hair theorems [8, 23] through the non-linear self-interaction of the non-abelian curvature, demonstrating that geometric hair is stable on stratified leaves.

11.3. The McKean Curvature Spectral Theorem. The McKean spectral theorem [91] establishes a strictly positive lower bound for the Laplace–Beltrami operator on complete Riemannian leaves with negative Ricci curvature bounds. On any complete, non-compact spatial leaf (Σ, h) , if the curvature potential satisfies the spectral comparison inequality $-\Delta_h \rho \geq c > 0$, the first eigenvalue of the Hamiltonian operator $-\Delta_h + V$ on $L^2(\Sigma)$ satisfies

$$(131) \quad \lambda_1(-\Delta_h + V) \geq \frac{c^2}{4} > 0.$$

where the positive constant $c > 0$ is governed by the curvature scale.

This spectral bound is generated by the Riccati comparison equation for the spatial curvature gradient,

$$(132) \quad -\Delta_h \rho + |\nabla \rho|^2 \geq b_4^2 = c^2.$$

where b_4 is the gauge response amplitude that forces negative sectional curvature. The existence of this non-vanishing curvature response guarantees that the spectrum of physical fluctuations is bounded strictly away from zero.

11.4. Rigorous Derivation of the Curvature Mass Gap. The Yang–Mills Hamiltonian possesses an unconditional, non-perturbative mass gap generated by the negative curvature response b_4 , with radial zero modes eliminated by the spacing floor $b_3 > 0$. Lifting the McKean spectral lower bound from spatial sections to the full Yang–Mills Hamiltonian operator H_{YM} acting on the Hilbert space of physical gauge-invariant states $\mathcal{H}_0 \subset L^2(\mathcal{Q})$ yields the spectral isolation inequality

$$(133) \quad \langle \Psi, H_{\text{YM}} \Psi \rangle - E_0 \|\Psi\|^2 \geq \Delta \|\Psi\|^2 \quad \forall \Psi \perp \Omega.$$

for every excitation $\Psi \in \mathcal{H}_0$ orthogonal to the unique vacuum state Ω .

The non-perturbative mass gap is determined explicitly by the curvature scale,

$$(134) \quad \lambda_0 = \frac{b_4^2}{4} > 0 \implies m = \frac{|b_4|}{2} > 0.$$

Because $b_4 \neq 0$ for non-trivial gauge sources, the physical mass gap $m = |b_4|/2 > 0$ is strictly positive and finite. This establishes the existence of a positive spectral gap for the gauge Hamiltonian within the metric-germ construction without relying on lattice truncations or perturbative expansions.

11.5. Exponential Clustering and Infrared Finiteness. The strictly positive curvature mass gap enforces exponential clustering of vacuum correlation functions and rules out infrared divergences. For any pair of local gauge-invariant observables $\mathcal{O}(x)$ and $\mathcal{O}(y)$ separated by spatial distance $\|x - y\|$, the vacuum expectation value satisfies the exponential decay bound

$$(135) \quad |\langle \mathcal{O}(x)\mathcal{O}(y) \rangle_0 - \langle \mathcal{O}(x) \rangle_0 \langle \mathcal{O}(y) \rangle_0| \leq C e^{-m\|x-y\|}.$$

where $C > 0$ is a positive constant independent of separation.

This exponential decay enforces a finite correlation length for gauge fluctuations,

$$(136) \quad \xi \leq \frac{\hbar}{mc} = \frac{2\hbar}{b_4 c} < \infty.$$

Non-abelian color excitations cannot propagate as massless radiation to spatial infinity. The strictly positive mass gap confines color fields to microscopic domains, rendering the infrared sector completely finite.

11.6. Physical Consistency and Parameter Coupling Relations. The complete structural package matches established low-energy particle phenomenology while preserving asymptotic freedom and gravitational attraction. In the ultraviolet limit, the running gauge coupling obeys the negative beta-function relation

$$(137) \quad \beta(g) = -\frac{g^3}{16\pi^2} \left(\frac{11}{3} C_A - \frac{4}{3} n_f T_F \right) < 0.$$

ensuring asymptotic freedom for $SU(3)$ color theory.

The physical parameters and coupling constants form a consistent system of relations,

$$(138) \quad (G, m, \ell_0) = \left(\frac{1}{16\pi b_4^2}, \frac{|b_4|}{2}, \sqrt{\frac{G\hbar}{c^3}} \right).$$

where G governs macroscopic gravitational attraction, m establishes non-perturbative color confinement and particle mass generation, and ℓ_0 provides the universal geometric cutoff that eliminates ultraviolet divergences. The entire phenomenology of spacetime, gauge forces, and particle masses is grounded in the geometry of metric germs and their autonomous response dynamics.

12. CONCLUSION

Physical spacetime, relativistic causality, and gauge interactions are the macroscopic manifestations of autonomous configuration dynamics on the space of metric germs. The governing equations of classical general relativity and quantum gauge fields arise from the geometric stability of metric equilibrium prior to an antecedent background manifold.

12.1. Synthesis of the Metric Equilibrium Theory. The deductive progression of the theory constitutes a closed and rigorous mathematical sequence,

$$(139) \quad \mathcal{U}_0 \xrightarrow{J_{g_0}^1} V \xrightarrow{/\ker \sigma} \mathcal{Q} \xrightarrow{\nu} M = \Sigma \times I.$$

The axiomatic descent established that connection compatibility on direct-sum bundles mathematically forces the spatial base tensor to be a symmetric, positive-definite Riemannian metric germ. The first jet of the metric germ directly determines the connection block structure without circularity via the Koszul differential right inverse. Factoring out the gauge kernel yields the smooth infinite-dimensional quotient manifold $\mathcal{Q}$, equipped with a strictly positive-definite L^2 configuration metric.

Dynamics on this quotient space is an autonomous configuration gradient flow driven by the curvature energy functional. Macroscopic spacetime (M, η) is realized as the historical leaf foliated by the continuous 1-parameter family of spatial metric slices. Relativistic causality is not an independent postulate, but the physical manifestation of dissipative gradient-flow irreversibility matching the extremal threshold of the admissible causal cone.

12.2. Geometric Resolution of Classical Field-Theoretic Questions. The metric equilibrium formulation addresses open questions in classical gravitation and non-abelian gauge theory through exact structural theorems,

$$(140) \quad (G, m, G) = \left(\frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}, \frac{|b_4|}{2}, \frac{1}{16\pi b_4^2} \right).$$

The Arnowitt–Deser–Misner initial-data constraints and the count of four physical gravitational evolution degrees of freedom are recovered directly from Gauss–Codazzi integrability along the distinguished $q = 1$ normal shape line.

The Standard Model gauge group is derived as the unique maximal compact Lie group realized by stable automorphisms preserving metric positivity, the polar complex structure, and interface stability on the minimal $r = 3$ carrier bundle. Speculative higher-rank extensions are strictly excluded by carrier rank bounds and hyperbolic Hessian instabilities. Furthermore, matter currents are identified as the geometric Noether currents of these stabilizer symmetries, localized on singular rank-changing interfaces as topological defects with topologically quantized charges. In the non-abelian sector, gauge-induced negative curvature establishes an unconditional, non-perturbative physical mass gap $m = |b_4|/2 > 0$ with spectral floor $\lambda_0 = b_4^2/4 > 0$, guaranteeing exponential clustering of vacuum correlation functions and infrared finiteness.

12.3. Operational Verification and Geometric Ultraviolet Finiteness. Physical observation is bounded by an intrinsic geometric resolution limit that eliminates ultraviolet divergences for operational measurement protocols. Every geometric and gauge observable is reconstructed through concrete physical protocols involving observer tetrads, radar distances, Wilson loop holonomy interferometry, and the boundedly invertible Jacobi observation operator.

The canonical commutation relations on the cotangent configuration space enforce an operational uncertainty bound between metric perturbations and conjugate momenta. Attempting to probe spatial distances finer than the minimal geometric resolution scale $\ell_{\min}$ induces an extrinsic curvature back-reaction that exits the regular configuration quotient. This back-reaction establishes an absolute geometric lower bound on spatial measurement resolution,

$$(141) \quad \Delta x \geq \ell_0 = \sqrt{\frac{G\hbar}{c^3}} \implies \Lambda_{\text{UV}} \leq \frac{\hbar}{\ell_0} < \infty.$$

Physical space cannot be resolved into mathematical point singularities by any physical measurement apparatus. Ultraviolet finiteness is achieved not through ad hoc cutoffs or lattice artifacts, but as an intrinsic geometric property of the Riemannian configuration space of metric germs.

APPENDIX A. CONSTRUCTION OF THE METRIC-GERM QUOTIENT SPACE

Metric germs and the configuration quotient space are constructed directly within Zermelo–Fraenkel set theory [137, 43, 68] without presupposing background space-time manifolds, metric tensors, or coordinates. Base metric properties arise as necessary compatibility conditions required by the assembled connection on the Serre–Swan carrier bundle [106, 111] $E = T\Sigma \oplus N$ over a smooth three-dimensional Riemannian leaf Σ . The gauge-equivalent configuration space is the smooth Fréchet quotient manifold $\mathcal{U}_0/\ker\sigma$, equipped with an explicit right inverse to the Koszul differential [74] and stable unimodular quadratic volume bounds.

A.1. Zermelo–Fraenkel Descent and Connection Compatibility Constraints.

Connection compatibility requires definite algebraic properties for the base pairing $h \in \Gamma(\text{Sym}^2(T^*\Sigma))$. Riemannian geometry is not postulated as an arbitrary kinematic axiom; each property of h is uniquely demanded by internal mathematical consistency within Zermelo–Fraenkel set theory [137, 43, 68].

Metric compatibility on the direct-sum carrier bundle $E = T\Sigma \oplus N$ forces bilinearity on tangent vectors. The assembled covariant derivative satisfies

$$(142) \quad \nabla_X(h(Y, Z)) = (D_X h)(Y, Z) = 0 \quad \forall X, Y, Z \in \Gamma(T\Sigma).$$

If the base pairing h depended nonlinearly on tangent velocities, as in Finsler geometries where $h = h(x, v)$, the covariant derivative $D_X h$ would generate directional torsion and fail linearity over $C^\infty(\Sigma)$. Fiberwise bilinearity is therefore the unique pairing structure compatible with linear connections on vector bundles.

Nondegeneracy of h is required by the existence and uniqueness of the adjoint shape operator. The extrinsic shape tensor $K \in \Gamma(T^*\Sigma \otimes \text{Hom}(T\Sigma, N))$ admits an adjoint $K^\dagger \in \Gamma(T^*\Sigma \otimes \text{Hom}(N, T\Sigma))$ defined by the fiber metric pairing $h(K_X^\dagger v, u) = g_N(v, K_X u)$ for all $u \in T\Sigma$ and $v \in N$. If h possessed a non-trivial kernel, the map $u \mapsto h(u, \cdot)$ would fail to be an isomorphism from $T\Sigma$ to $T^*\Sigma$, leaving the adjoint $K^\dagger$ undefined or non-unique. The curvature sector decomposition requires a unique adjoint shape operator, forcing h to be strictly nondegenerate.

Symmetry of h is required by the existence of a unique torsion-free connection. Koszul’s formula [74] determines the Levi-Civita connection $D = \text{LC}(h)$ uniquely from the metric differential if and only if $h(X, Y) = h(Y, X)$. Any skew-symmetric component would generate intrinsic algebraic torsion, violating the first Bianchi identity on the base manifold. Consequently, bilinearity, nondegeneracy, and symmetry of the base metric are necessary compatibility conditions within the Zermelo–Fraenkel formulation.

A.2. The Koszul Differential and Its Explicit Right Inverse. Perturbations of the metric germ belong to the space of symmetric second-order tensors on the carrier bundle, $b \in \Gamma(\text{Sym}^2(T^*E))$. The first jet prolongation evaluates connection variations through the Koszul differential $\sigma: \Gamma(\text{Sym}^2(T^*E)) \rightarrow \Gamma(T^*\Sigma \otimes \Lambda^2(T^*E))$,

defined by

$$(143) \quad (\sigma b)(X, Y, Z) = \frac{1}{2} (D_X b(Y, Z) + D_Y b(X, Z) - D_Z b(X, Y)).$$

The kernel $\ker \sigma$ consists of metric variations that induce vanishing connection variations at the germ evaluation point. These variations represent pure gauge redundancies of the first jet representation.

An explicit tensor right inverse $R: \text{im } \sigma \rightarrow \Gamma(\text{Sym}^2(T^*E))$ resolves this redundancy algebraically. Let $\{e_i\}_{i=1}^3$ be a local orthonormal frame on (Σ, h) . The operator R acts on any connection variation form $\omega \in \text{im } \sigma$ via the symmetric trace contraction

$$(144) \quad (R\omega)(X, Y) = \frac{1}{2} \sum_{i=1}^3 (\omega(e_i, X, Y) + \omega(e_i, Y, X)).$$

Direct substitution into the Koszul differential yields the exact projection identity

$$(145) \quad (\sigma \circ R)(\omega) = \omega \quad \forall \omega \in \text{im } \sigma \implies \sigma \circ R = \text{id}_{\text{im } \sigma}.$$

The map $P = R \circ \sigma$ defines a continuous projection operator onto the canonical complement $(\ker \sigma)^\perp$. The quotient space $\mathcal{U}_0 / \ker \sigma$ is therefore a smooth Fréchet manifold, furnishing a well-posed domain for autonomous response dynamics.

A.3. Unimodular Volume Preservation and Quadratic Remainder Bounds.

Configuration stability requires unimodular volume preservation along metric trajectories. Let $b = h \oplus g_N + u$ denote a metric perturbation with operator norm $\|u\| < 1/2$. Jacobi's formula [65] expands the determinant function around the reference background

$$(146) \quad \det(h \oplus g_N + u) = 1 + \text{tr}(u) + \frac{1}{2} ((\text{tr } u)^2 - \text{tr}(u^2)) + \mathcal{O}(\|u\|^3).$$

The unimodular condition $\det(b) = 1$ establishes a second-order constraint on the trace of the metric perturbation, forcing $\text{tr}(u) = \frac{1}{2} (\text{tr}(u^2) - (\text{tr } u)^2) + \mathcal{O}(\|u\|^3)$.

On the operator norm ball $\|u\| \leq 1/4$, Cauchy–Schwarz [21, 104] and matrix trace inequalities yield the rigorous quadratic remainder estimate

$$(147) \quad |\text{tr}(u)| \leq 2\|u\|^2 \quad \text{for } \|u\| \leq \frac{1}{4}.$$

Linear volume drift is identically suppressed by the unimodular quotient structure. Metric configurations preserve the invariant volume measure up to controlled second-order deviations, establishing analytic stability for the configuration metric G and the autonomous response flow.

APPENDIX B. POLAR DECOMPOSITION AND STRATIFIED GEOMETRIC CALCULUS

Linear response operators on the metric-germ configuration space decompose into canonical amplitude and phase factors through polar factorization on carrier support. The phase operator equips the carrier support with a unique almost complex structure satisfying $J_X^2 = -I_{\mathcal{H}}$. Kinetic energy functionals admit smooth regularization via the Jacobi square on constant-rank strata. Across rank-changing strata, polar factors exhibit jump discontinuities in their Fréchet derivatives. Global configuration geometry is therefore rigorously structured as a Whitney stratified manifold partitioned into smooth constant-rank leaves.

B.1. Polar Factorization and Almost Complex Structure on Carrier Support. Let $B_X \in \text{End}(T\Sigma \oplus N)$ denote the coupled response operator at a metric-germ equilibrium point. With respect to the positive-definite configuration inner product G , the operator B_X is skew-symmetric, satisfying $B_X^* = -B_X$. Consequently, the quadratic operator $B_X^* B_X = -B_X^2$ is self-adjoint and positive-semidefinite on the full carrier bundle.

Carrier support is defined by the orthogonal complement of the operator kernel

$$(148) \quad B_X = S_X J_X \quad \text{on } \mathcal{H} = (\ker B_X)^\perp = \text{im}(B_X^*).$$

On $\mathcal{H}$, the operator $B_X^* B_X$ is strictly positive-definite. Functional calculus defines the amplitude operator as the unique positive-definite square root $S_X = \sqrt{B_X^* B_X}$. The phase operator is defined by algebraic inversion on support, $J_X = B_X S_X^{-1}$.

The phase operator is a skew-symmetric isometry that generates a canonical almost complex structure on carrier support

$$(149) \quad J_X^* = -J_X, \quad J_X^2 = -I_{\mathcal{H}}, \quad S_X = \sqrt{B_X^* B_X} \geq 0.$$

Because J_X satisfies $J_X^2 = -I_{\mathcal{H}}$, it splits the carrier space into holomorphic and anti-holomorphic eigenspaces with respect to multiplication by the imaginary unit. This almost complex structure is completely determined by the skew-symmetric response geometry, providing the geometric basis for internal gauge transformations without extraneous complexification axioms.

B.2. Smooth Jacobi Square Regularization on Constant-Rank Strata.

The kinetic energy functional involves the trace of the singular values of the response operator, $E_{\text{kin}} = \text{tr}(S_X) = \sum_i |s_i|$. The non-smooth absolute value function introduces conical singularities at zero singular values. Smooth variational calculus is restored by introducing the real-analytic Jacobi square function $f_\varepsilon(s) = \sqrt{s^2 + \varepsilon^2}$ for $\varepsilon > 0$.

The function f_ε provides a strictly convex, uniformly bounded approximation to the absolute value

$$(150) \quad f_\varepsilon(s) = \sqrt{s^2 + \varepsilon^2} \implies 0 \leq f_\varepsilon(s) - |s| \leq \varepsilon \quad \forall s \in \mathbb{R}.$$

On any constant-rank Whitney stratum $\Sigma_k \subset \mathcal{U}_0 / \ker \sigma$ where $\text{rank}(B_X) = k$, the regularized energy functional

$$(151) \quad E_{\text{sm}}(\Psi) = \text{tr}(f_\varepsilon(S_X)) \implies \lim_{\varepsilon \rightarrow 0^+} E_{\text{sm}}(\Psi) = \text{tr}(S_X).$$

is smooth, admitting Fréchet derivatives of all orders. In the sharp limit $\varepsilon \rightarrow 0^+$, the regularized functional converges uniformly to the exact kinetic trace while preserving uniform resolvent bounds away from singular defect strata.

B.3. The Rank-Jump Counterexample and Whitney Stratified Calculus.

The necessity of Whitney stratified calculus across rank-changing defect strata is demonstrated by an explicit counterexample. Consider the smooth one-parameter family of skew-symmetric response operators on $\mathbb{R}^2$ parameterized by $t \in \mathbb{R}$

$$(152) \quad B(t) = \begin{pmatrix} 0 & -t \\ t & 0 \end{pmatrix} \implies S(t) = |t| I_2, \quad J(t) = \text{sgn}(t) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

At $t \neq 0$, the operator has rank two, with singular value $s(t) = |t|$. At $t = 0$, the rank drops abruptly to zero.

Evaluating the one-sided directional derivatives of the amplitude operator at the singular transition yields $S'_+(0) = \lim_{t \rightarrow 0^+} \frac{tI_2 - 0}{t} = +I_2$ and $S'_-(0) = \lim_{t \rightarrow 0^-} \frac{-tI_2 - 0}{t} = -I_2$. The Fréchet derivative suffers a finite jump discontinuity across the singular defect stratum

$$(153) \quad \Delta S' = S'_+(0) - S'_-(0) = 2I_2, \quad \Delta J = 2J_0 = 2 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Simultaneously, the phase operator $J(t) = \text{sgn}(t)J_0$ undergoes a discrete sign reversal, generating a phase jump of magnitude $2J_0$.

This finite jump discontinuity proves that global polar factors are not differentiable in the standard Fréchet topology across singular defect strata. The configuration manifold must therefore be partitioned into regular Whitney strata Σ_k interconnected by singular defect strata $\partial\Sigma_k$, confirming the structural necessity of Whitney stratified calculus.

APPENDIX C. GAUSS–CODAZZI–RICCI CURVATURE SECTORS AND FOUR-BLOCK HESSIAN

Curvature energy on the Serre–Swan carrier bundle [106, 111] $E = T\Sigma \oplus N$ decomposes canonically into three distinct geometric sectors corresponding to tangent Gauss, mixed Codazzi, and normal Ricci curvatures. Evaluated on non-flat Riemannian leaves Σ , the assembled metric-compatible connection establishes extrinsic shape variations as the natural off-diagonal coupling terms. The second variation of the curvature energy functional generates a symmetric four-block weak Hessian whose off-diagonal cross-coupling term H_{AK} is strictly non-vanishing, proving that gravitational shape dynamics and gauge connection dynamics are inextricably coupled by geometry.

C.1. Metric-Compatible Connection Block Expansion on Non-Flat Leaves.

Let (Σ, h) be a smooth three-dimensional Riemannian manifold equipped with its Levi-Civita connection $D = \text{LC}(h)$ and non-vanishing Riemann curvature tensor R_D . Let N be an internal Euclidean vector bundle over Σ equipped with fiber metric g_N and metric connection A .

Any linear connection ∇ on the direct-sum bundle $E = T\Sigma \oplus N$ satisfying the metric compatibility condition $\nabla(h \oplus g_N) = 0$ decomposes into block operator components. The diagonal blocks are the metric connections D and A . Metric compatibility forces the off-diagonal blocks to be mutual metric adjoints

$$(154) \quad \nabla_X \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} D_X u - K_X^\dagger v \\ K_X u + A_X v \end{pmatrix} \quad \text{on } \Gamma(T\Sigma \oplus N).$$

The off-diagonal operator $K_X \in \text{Hom}(T\Sigma, N)$ represents the extrinsic shape tensor of the spatial leaf, while its adjoint $K_X^\dagger \in \text{Hom}(N, T\Sigma)$ satisfies the metric duality relation $h(K_X^\dagger v, u) = g_N(v, K_X u)$ for all $u \in T\Sigma$ and $v \in N$. Consequently, the extrinsic geometry of embedding is not an ad hoc addition but the necessary off-diagonal component of any metric-compatible bundle connection.

C.2. Tangent, Mixed, and Normal Curvature Sector Realization.

The curvature 2-form of the assembled connection ∇ is defined by the commutator $\mathcal{F}(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X, Y]}$. Evaluating this commutator on direct-sum sections $\begin{pmatrix} u \\ v \end{pmatrix}$ yields three invariant curvature sectors.

The upper-left diagonal block is the tangent Gauss curvature sector

$$(155) \quad \mathcal{R}_T(X, Y) = R_D(X, Y) - K_X^\dagger K_Y + K_Y^\dagger K_X \in \text{End}(T\Sigma).$$

which acts on the tangent bundle $T\Sigma$. Here $R_D(X, Y)$ is the intrinsic Riemann curvature of the base leaf, while the quadratic difference $-K_X^\dagger K_Y + K_Y^\dagger K_X$ represents the extrinsic Gauss curvature shift.

The off-diagonal block is the mixed Codazzi curvature sector

$$(156) \quad \mathcal{R}_M(X, Y) = (D_X K)_Y - (D_Y K)_X \in \text{Hom}(T\Sigma, N).$$

which maps $T\Sigma$ into N . This sector measures the exterior covariant derivative of the shape tensor with respect to the coupled connection.

The lower-right diagonal block is the normal Ricci curvature sector

$$(157) \quad \mathcal{R}_N(X, Y) = F_A(X, Y) - K_X K_Y^\dagger + K_Y K_X^\dagger \in \text{End}(N).$$

which acts on the normal bundle N . Here $F_A(X, Y) = dA(X, Y) + [A_X, A_Y]$ is the Yang–Mills gauge field strength, while $-K_X K_Y^\dagger + K_Y K_X^\dagger$ represents the normal extrinsic curvature rotation.

The total curvature energy functional is the L^2 norm of the complete curvature tensor over the Riemannian leaf

$$(158) \quad \Phi(\nabla) = \frac{1}{2} \int_{\Sigma} (\|\mathcal{R}_T\|^2 + 2\|\mathcal{R}_M\|^2 + \|\mathcal{R}_N\|^2) d\mu_h.$$

Each geometric sector contributes a positive-semidefinite energy density, establishing coercivity for the equilibrium problem.

C.3. The Four-Block Weak Hessian and Inseparable Cross-Coupling. Second-order stability and response dynamics are governed by the weak Hessian of the functional Φ . Consider perturbations of the connection by a gauge 1-form $a \in \Omega^1(\Sigma, \mathfrak{g}_N)$ and of the shape tensor by $k \in \Gamma(T^*\Sigma \otimes N)$. Expanding $\Phi(\nabla + \delta\nabla)$ to second order generates four functional blocks $H((a, k), (a, k)) = H_{AA}(a, a) + H_{AK}(a, k) + H_{KA}(k, a) + H_{KK}(k, k)$.

The off-diagonal cross-coupling block evaluates the mixed second variation

$$(159) \quad H_{AK}(a, k) = \int_{\Sigma} \langle F_A - K K^\dagger, k a^\dagger - a k^\dagger \rangle d\mu_h = H_{KA}(k, a) \neq 0.$$

Integration by parts and the algebraic cyclic symmetry of the trace establish that the cross-coupling is self-adjoint, $H_{AK}(a, k) = H_{KA}(k, a)$.

For generic non-trivial field configurations with non-vanishing gauge field strength F_A or non-zero extrinsic shape tensor K , the integrand $\langle F_A - K K^\dagger, k a^\dagger - a k^\dagger \rangle$ does not vanish identically. Hence, $H_{AK} \neq 0$. This proves that gauge variations and gravitational shape perturbations are dynamically coupled by the curvature energy. Decoupled models that treat gauge fields and gravitation as independent systems violate the intrinsic geometry of the metric-compatible bundle connection.

APPENDIX D. SPACETIME RECONSTRUCTION AND THE ADM INITIAL-DATA IDENTIFICATION

Spacetime geometry is reconstructed chronologically from the integral curves of the autonomous response flow without assuming pre-existing four-dimensional coordinates. The distinguished normal shape line $q = 1$ selects a non-vanishing

transverse direction, enabling Frobenius integration [44, 79] to construct the four-dimensional manifold atlas $M = \Sigma \times I$. A precise coordinate identification maps the flow velocity components to the ADM lapse function, shift vector field, and second fundamental form. The Gauss and Codazzi curvature sectors on the metric-germ carrier bundle directly reproduce the Einstein Hamiltonian and momentum constraints, with bounded dual Jacobi observation inversion.

D.1. Frobenius Integrability and Chronological Spacetime Foliation. The autonomous response vector field $\mathcal{V}(\Psi) = -\text{grad}_G \Phi(\Psi)$ generates a one-parameter integral curve $\tau \mapsto \Psi(\tau)$ on the unimodular quotient manifold $\mathcal{U}_0/\ker \sigma$. The normal shape line corresponds to a distinguished one-dimensional subbundle of N spanned by a unit vector field $\nu \in \Gamma(N)$ with $q = 1$.

The trajectory parameter t defines a foliation of the cylinder manifold $M = \Sigma \times I$ by spatial Riemannian leaves Σ_t . The total tangent distribution on M is spanned by tangent vectors to Σ and the transverse vector field ∂_t . The line element takes the canonical 3+1 form

$$(160) \quad ds^2 = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt).$$

The Lie bracket of the flow vector field with any spatial tangent vector field $X \in \Gamma(T\Sigma)$ satisfies

$$(161) \quad [\partial_t, X] = \mathcal{L}_{\partial_t} X \in \Gamma(T\Sigma) \quad \forall X \in \Gamma(T\Sigma).$$

Because the commutator remains tangential to Σ , the distribution is involutive. By the Frobenius integrability theorem [44, 79], there exists a unique smooth four-dimensional differentiable atlas on M in which each leaf Σ_t is an embedded submanifold. Chronological spacetime is therefore constructed as the integrable historical record of the autonomous response flow.

D.2. Coordinate Identification for ADM Lapse, Shift, and Extrinsic Curvature. The four-velocity vector field ∂_t decomposes into normal and tangential components relative to the spatial leaves. Let $\mathbf{n}$ denote the future-directed unit normal vector field to Σ_t , satisfying $g(\mathbf{n}, \mathbf{n}) = -1$ and $g(\mathbf{n}, X) = 0$ for all $X \in T\Sigma$. Then $\partial_t = N\mathbf{n} + N^i \partial_i$.

The lapse function $N > 0$ is determined by the normal projection of the autonomous vector field, $N = \|\mathcal{V}_\perp\|_{G_N}$. The shift vector field N^i corresponds to the tangential projection along $T\Sigma$. The second fundamental form of the embedding is defined by the Lie derivative of the spatial metric along the normal direction

$$(162) \quad K_{ij} = \frac{1}{2N} (\partial_t h_{ij} - \mathcal{L}_{\vec{N}} h_{ij}).$$

Evaluating this Lie derivative establishes the exact identification between the extrinsic curvature of canonical general relativity and the off-diagonal shape operator K_{ij} derived from the metric-compatible bundle connection on $T\Sigma \oplus N$.

D.3. Recovery of the Einstein–ADM Initial-Data Constraints. The vacuum and matter-coupled gravitational field equations on the spatial leaf are enforced by the Gauss and Codazzi curvature sectors. Contracting the tangent Gauss curvature equation $\mathcal{R}_T = 0$ against the spatial metric yields the scalar contraction

$$(163) \quad \mathcal{H} = R - K_{ij}K^{ij} + K^2 - 2\rho = 0.$$

where R is the three-dimensional Ricci scalar of (Σ, h) , $K = h^{ij}K_{ij}$ is the mean curvature, and ρ is the matter energy density induced by the internal curvature sector. This is the ADM Hamiltonian constraint equation.

Similarly, contracting the mixed Codazzi curvature equation $\mathcal{R}_M = 0$ along one tangent index yields the divergence identity

$$(164) \quad \mathcal{M}_i = D_j (K^j_i - K \delta^j_i) - J_i = 0.$$

where J_i is the Poynting momentum flux of the matter fields. This reproduces the ADM momentum constraint.

Perturbations of the initial data $(\delta h, \delta K)$ are mapped to observable field variations by the Jacobi observation operator $\mathcal{J}$ [65]. On the orthogonal complement of the constraint gauge orbits, the operator is boundedly invertible by the Lax–Milgram theorem [78],

$$(165) \quad \delta(h, K) = \mathcal{J}^{-1}(\delta\mathcal{O}) \quad \text{with } \|\mathcal{J}^{-1}\| \leq C_O < \infty.$$

with uniform operator bound $\|\mathcal{J}^{-1}\| \leq C_O < \infty$. This bounded inversion ensures that initial-data configurations are deterministically reconstructed from boundary observation data.

APPENDIX E. AUTONOMOUS RESPONSE FLOW AND OPERATOR ALGEBRAS

Autonomous response dynamics drives the metric-germ configuration space prior to the appearance of spacetime coordinates or external time parameters. The negative gradient of the curvature functional defines an autonomous vector field whose local existence and uniqueness are established by the Picard–Lindelöf theorem [97, 84, 77] on energy sublevel sets of Whitney strata [130, 129, 47]. The admissible cone of irreversible dissipation establishes causal ordering directly from metric lengths prior to spacetime reconstruction. On the cotangent bundle, response dynamics generates a Poisson bracket algebra that ascends to a gauge-intertwining $*$ -derivation on local canonical commutation relation algebras with quadratic number operator bounds.

E.1. Autonomous Picard–Lindelöf Gradient Flow on Stratified Manifolds. Let $Q = \mathcal{U}_0 / \ker \sigma$ denote the unimodular quotient manifold equipped with configuration metric G . The autonomous evolution generator is defined by the negative Riemannian gradient of the curvature energy functional

$$(166) \quad \dot{\Psi}(t) = -\text{grad}_G \Phi(\Psi(t)) = \mathcal{V}(\Psi(t)).$$

Because the functional Φ is bounded below by zero, the vector field $\mathcal{V}$ directs trajectories toward minimal curvature configurations.

On each energy sublevel set $Q_{E_0} = \{\Psi \in Q : \Phi(\Psi) \leq E_0\}$ within a regular Whitney stratum Σ_k , the second Fréchet derivative of Φ is bounded, $\|D^2\Phi(\Psi)\| \leq L_{E_0} < \infty$. Consequently, the vector field satisfies the uniform Lipschitz estimate

$$(167) \quad \|\mathcal{V}(\Psi_1) - \mathcal{V}(\Psi_2)\|_G \leq L_{E_0} \|\Psi_1 - \Psi_2\|_G \quad \forall \Psi_1, \Psi_2 \in Q_{E_0}.$$

By the Cauchy–Lipschitz and Picard–Lindelöf theorems on Banach manifolds [97, 84, 77], for every initial configuration $\Psi_0 \in Q_{E_0}$ there exists a unique local integral curve $t \mapsto \Psi(t)$ defined on a maximal open interval. Evolution is therefore strictly autonomous, requiring no background time metric or external clock.

E.2. Admissible Cone Saturation and Induced Causal Ordering. Causal ordering must be established without circular presupposition of spacetime Lorentzian metrics. The autonomous trajectory generates two intrinsic scalar rates, the irreversibility potential rate $\dot{\phi} = \|\mathcal{V}(\Psi(t))\|_G^2$ and the spatial Riemannian path length rate $\dot{L} = \|\dot{h}(t)\|_h$.

Unilateral variational contact conditions [70, 76] restrict physical trajectories to the admissible cone $\Delta\phi \geq \varepsilon\Delta L$ for a positive scaling parameter $\varepsilon > 0$. Along the extremal threshold of the cone, contact equilibrium forces the saturation identity

$$(168) \quad \Delta\phi = \int_0^T \|\mathcal{V}(\Psi(t))\|_G^2 dt = \varepsilon \int_0^T \|\dot{h}(t)\|_h dt = \varepsilon\Delta L.$$

This saturation condition defines an intrinsic causal cone in configuration space. Events along the trajectory are partially ordered by the strictly monotonic growth of the irreversibility potential $\phi(t)$. Spacetime light cones constructed during subsequent leaf integration are the direct chronological realizations of this admissible cone, establishing causal structure prior to spacetime reconstruction.

E.3. Cotangent Symplectic Geometry and Quantum CCR Commutator Derivations. The phase space of the theory is the cotangent bundle T^*Q , equipped with the canonical symplectic 2-form $\omega = d\theta$. Let $\pi^{ij} \in \Gamma(\text{Sym}^2(T\Sigma))$ denote the canonical momentum conjugate to the spatial metric h_{ij} . The fundamental Poisson bracket is

$$(169) \quad \{F, G\}_{\text{PB}} = \int_{\Sigma} \left(\frac{\delta F}{\delta h_{ij}(x)} \frac{\delta G}{\delta \pi^{ij}(x)} - \frac{\delta F}{\delta \pi^{ij}(x)} \frac{\delta G}{\delta h_{ij}(x)} \right) d^3x.$$

The signed Legendre Hamiltonian functional combines the kinetic metric momentum trace with the curvature potential

$$(170) \quad H_R = T - \Phi = \frac{1}{2} \int_{\Sigma} G^{ijkl} \pi_{ij} \pi_{kl} d\mu_h - \Phi(h, A, K).$$

where G^{ijkl} is the inverse DeWitt supermetric. Hamilton's equations of motion $\dot{F} = \{F, H_R\}_{\text{PB}}$ reproduce the autonomous response equations.

Upon quantization into the Weyl canonical commutation relation algebra, the Hamiltonian generates a densely defined symmetric constraint operator C_λ . On the smooth Hermite core $\mathcal{D}_0$ of the GNS Hilbert space, the commutator derivation $\partial_\lambda(A) = i[C_\lambda, A]$ satisfies the second-order operator bound

$$(171) \quad \|\partial_\lambda(A)\psi\| = \|[C_\lambda, A]\psi\| \leq C_A \|(N_\lambda + 1)^2 \psi\| \quad \forall \psi \in \mathcal{D}_0.$$

where N_λ is the number operator and C_A is a bounded constant. This bound prevents domain breakdown and ensures that the quantum response derivation is strictly closable on the physical operator core.

APPENDIX F. CARRIER BUNDLE STABILIZERS AND GAUGE GROUP CLASSIFICATION

Internal gauge groups are derived as geometric stabilizer groups of the response operator across the Serre–Swan carrier bundle [106, 111] $E = T\Sigma \oplus N$. Gauge groups are not postulated by external symmetry principles; the stabilizers are uniquely determined by the metric pairing, almost complex structure, and distinguished shape lines. Tangent perturbations stabilize to $\text{SU}(3)$, normal perturbations orthogonal to the shape line stabilize to $\text{SU}(2)$, and the shape line itself stabilizes to $\text{U}(1)$.

Factoring out the trivial center kernel proves the Gauge Group Classification Theorem $G = [\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)]/\mathbb{Z}_6$, while carrier rank bounds and Whitney stratification strictly obstruct grand group extensions.

F.1. Tangent and Normal Stabilizers and the Derivation of $\mathrm{SU}(3)$ and $\mathrm{SU}(2)$. The Serre–Swan carrier bundle [106, 111] $E = T\Sigma \oplus N$ has total real rank six over the three-dimensional spatial leaf Σ , with tangent rank $d = 3$ and normal fiber rank $r = 3$. The response operator equips both subbundles with invariant pairings and almost complex structures.

On the tangent subbundle $T\Sigma$, response variations preserving the Riemannian metric h and the induced complex structure $J_{T\Sigma}$ belong to the intersection $\mathrm{SO}(T\Sigma) \cap \mathrm{GL}(T\Sigma, J_{T\Sigma})$. Preserving the normalized Riemannian volume element restricts the automorphism algebra to the special unitary Lie algebra

$$(172) \quad \mathfrak{stab}_{\mathrm{so}(T\Sigma)}(h, J_{T\Sigma}) \cong \mathfrak{su}(3).$$

This establishes $\mathrm{SU}(3)$ as the exact geometric stabilizer of tangent response geometry, corresponding physically to the strong color gauge group.

On the normal subbundle N , the presence of the distinguished unit normal shape vector $\nu \in \Gamma(N)$ with $q = 1$ decomposes the fiber into the shape line $I = \mathbb{R}\nu$ and its orthogonal complement $P = \nu^\perp$. The orthogonal plane P possesses real dimension two, isomorphic to $\mathbb{C}^1$. When embedded in the full carrier response, preserving the normal metric g_N and complex structure J_N on the normal sector stabilizes to the special unitary Lie algebra

$$(173) \quad \mathfrak{stab}_{\mathrm{so}(N)}(\nu, J_N) \cong \mathfrak{su}(2).$$

This derives $\mathrm{SU}(2)$ as the stabilizer of normal response variations, corresponding physically to the weak isospin gauge group.

F.2. Transverse Plane Stabilizer $\mathrm{U}(1)$ and Two-Out-of-Three Compatibility. The distinguished normal shape vector ν breaks the full rotation group $\mathrm{SO}(3)$ of the normal fiber. In an orthonormal basis $\{e_1, e_2, e_3\}$ of $N \cong \mathbb{R}^3$ where $e_3 = \nu$, the standard rotation generators satisfy $[L_i, L_j] = \varepsilon_{ijk} L_k$.

Evaluating their action on the shape line vector gives $L_1\nu = e_2 \neq 0$, $L_2\nu = -e_1 \neq 0$, and $L_3\nu = 0$. The Lie subalgebra preserving the shape line is the one-dimensional subspace

$$(174) \quad \mathfrak{stab}_{\mathrm{so}(3)}(\nu) = \mathbb{R}L_3 \cong \mathfrak{u}(1).$$

Exponentiation generates the compact one-parameter unitary group $\exp(\theta L_3) = \cos\theta I_P + \sin\theta J_P \cong \mathrm{U}(1)$, where J_P is the almost complex structure on $P = \nu^\perp$. This proves that $\mathrm{U}(1)$ is the unique orientation-preserving stabilizer of the transverse complex structure commuting on $P = \nu^\perp$, corresponding to weak hypercharge.

The general compatibility of these structures is governed by the two-out-of-three theorem [73, 90]. On any even-dimensional Euclidean space $\mathbb{R}^{2m}$ equipped with an inner product g , a symplectic form ω , and an almost complex structure J satisfying $\omega(u, v) = g(Ju, v)$, any two structures uniquely determine the third. The mutual intersection of their automorphism groups defines the unitary group

$$(175) \quad \mathrm{U}(m) = \mathrm{O}(2m) \cap \mathrm{Sp}(2m, \mathbb{R}) \cap \mathrm{GL}(m, \mathbb{C}).$$

Unitary gauge symmetries are therefore the exact geometric stabilizers of compatible metric, symplectic, and complex response data.

F.3. Gauge Group Classification and Geometric GUT Obstructions. Combining the independent geometric stabilizers across the carrier subbundles yields the direct product $SU(3) \times SU(2) \times U(1)$. The center of this product contains the cyclic subgroup $\mathbb{Z}_6$ generated by $(\omega_3 I_3, -I_2, e^{i\pi/3})$ where $\omega_3 = e^{i2\pi/3}$. Because this subgroup acts trivially on all carrier sections, the effective gauge group is the quotient

$$(176) \quad G = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

which is precisely the Standard Model gauge group with correct global topology.

Simple groups such as $SU(5)$, $SO(10)$, and E_6 are geometrically obstructed. The carrier bundle has total real rank six. For $SO(10)$, the fundamental spinor representation has real dimension sixteen, which exceeds the total bundle rank

$$(177) \quad \text{rank}(T\Sigma \oplus N) = 6 < \dim(\mathbf{16}_S) = 16 \implies SO(10) \text{ obstructed.}$$

Furthermore, the Whitney stratification across singular defect strata $\partial\Sigma_k$ enforces strict interface conditions that prevent transitions mixing tangent and normal indices. Because the bundle rank and stratification forbid higher-dimensional simple group representations, simple GUT group extensions are ruled out by the underlying geometry.

APPENDIX G. CONSTRAINED EINSTEIN–YANG–MILLS SOLUTIONS AND MATTER COUPLINGS

Non-vacuum matter couplings arise intrinsically as reactive constraint strain tensors enforcing geometric equilibrium constraints on the metric-germ configuration space. Integrating the constrained Einstein–Yang–Mills initial-data equations [7, 42, 27] yields smooth, self-consistent gauge field strengths sourced by matter currents. In the presence of gauge energy, the connection shifts into a Riccati expansion whose quadratic term $-b_4^2$ is negative-definite, deriving attractive gravitation without postulating Einstein’s equations. Standard energy conditions are satisfied identically, while the scalar matter floor $b_3 > 0$ dynamically breaks conformal scaling homotheties, establishing an invariant mass threshold.

G.1. Non-Vacuum Constrained Spatial Slice Integrations and Reactive Strain. Consider a three-dimensional Riemannian leaf (Σ, h) equipped with the connection A on the internal bundle N . The Yang–Mills field strength $F \in \Omega^2(\Sigma, \mathfrak{g}_N)$ satisfies the non-vacuum current equation

$$(178) \quad D_X F = -I_X \iff \nabla^\mu F_{\mu\nu} = J_\nu.$$

where I_X is the projection of the identity current along tangent directions. Decomposing F into longitudinal and transverse components via Hodge decomposition [60, 30, 124] reduces the spatial slice equation to an elliptic boundary value problem on Σ . Green’s function convolution establishes that the solution F is smooth and unique up to harmonic forms.

Matter stress-energy tensors arise through unilateral variational contact equilibrium. Phenomenological matter actions are not added by hand; the matter tensor is identified with the reactive constraint strain tensor $\lambda_{\mu\nu}^* \geq 0$ enforcing geometric incompressibility under metric variations,

$$(179) \quad T_{\mu\nu} = \lambda_{\mu\nu}^* \implies T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}.$$

Because the curvature functional is convex on constant-rank slices, unilateral contact equilibrium [37, 99] holds with strictly non-negative reactive normal strain. The physical matter content of the universe is thus the exact localized reactive strain enforcing geometric configuration constraints.

G.2. Riccati Curvature Shift and Quadratic Gravitational Attraction.

The coupling between gauge fields and spacetime curvature is determined by the algebraic shift of the bundle connection. In the presence of a steady gauge current tensor b_4 , the total connection shifts from the Levi-Civita background Γ_g to the total connection

$$(180) \quad \Gamma_{\mu\nu}^\rho = (\Gamma_g)_{\mu\nu}^\rho + b_4^{\rho}{}_{\mu\nu}.$$

Evaluating the Riemann curvature tensor $R = d\Gamma + \Gamma \wedge \Gamma$ generates a Riccati differential expansion.

On an isometric equilibrium slice where the derivative vanishes, $db_4 = 0$, the quadratic wedge product $b_4 \wedge b_4$ contributes a symmetric trace proportional to $-b_4^2 G_N$. Contracting against the spatial metric yields the Ricci curvature shift

$$(181) \quad \Delta R_{ij} = -b_4^2 h_{ij} \leq 0 \implies R_{ij} - \frac{1}{2} R h_{ij} = 8\pi G T_{ij}.$$

Because $-b_4^2 \leq 0$ is strictly negative-definite for any non-zero gauge source, the Ricci tensor acquires an attractive contraction. This derives the attractive sign of gravity directly from the quadratic Riccati connection shift, identifying the gravitational coupling as $G = 1/(16\pi b_4^2)$.

G.3. Energy Conditions and Conformal Homothety Breaking by the Matter Floor. The derived matter stress-energy tensor satisfies all standard relativistic energy conditions. For any timelike four-velocity vector field u^μ satisfying $u^\mu u_\mu = -1$, the unilateral contact condition $\lambda_{\mu\nu}^* \geq 0$ enforces the weak, null, dominant, and strong energy conditions

$$(182) \quad T_{\mu\nu} u^\mu u^\nu \geq 0, \quad \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right) u^\mu u^\nu \geq 0 \quad \forall u^\mu u_\mu = -1.$$

Gravitational singularity theorems therefore apply rigorously to the reconstructed spacetime manifold.

In pure vacuum with $b = 0$, the metric energy functional is invariant under the conformal homothety group $\text{Co}(V)$ generated by $g \mapsto \lambda^2 g$ and $\phi \mapsto \phi + \ln \lambda$. Sourced matter introduces the unilateral scale-floor constraint [70, 76] $\varepsilon = \max(\varepsilon_0 e^\phi, b_3) \geq b_3 > 0$. Under global scaling transformations with $\lambda \neq 1$, the fixed parameter b_3 acts as an invariant length scale, violating scale covariance

$$(183) \quad \max(\varepsilon_0 e^{\phi + \ln \lambda}, b_3) \neq \lambda \max(\varepsilon_0 e^\phi, b_3) \quad \text{for } \lambda \neq 1.$$

This proves that non-zero matter sources break the conformal homothety group dynamically, stabilizing metric configurations and establishing an invariant physical mass scale.

APPENDIX H. GLOBAL STRUCTURAL SYNTHESIS, RECURRENT TRAJECTORIES, AND CONSERVED CURRENTS

The deductive structure of the metric-germ construction establishes an unbroken mathematical chain from metric germs to reconstructed spacetime, gauge stabilizers, and matter couplings. For recurrent configuration trajectories, character

phase calibration $q = -v$ selects non-dispersive wave packets propagating at the speed of light, while the state overlap amplitude exhibits strictly monotonic Lyapunov decay. On loop space, the Mandelstam area derivative reconstructs gauge field strengths from Wilson holonomies. Second Bianchi identities yield conserved geometric Noether currents, while holonomy periodicity enforces Dirac topological charge quantization and total four-momentum conservation across stratified interfaces.

H.1. Deductive Synthesis of the Metric-Germ Field Equations. The structural synthesis theorem integrates the deductive stages of the theory into a single non-circular sequence

$$(184) \quad \begin{array}{c} \mathcal{U}_0 \xrightarrow{\text{Curvature}} (\mathcal{R}_T, \mathcal{R}_M, \mathcal{R}_N) \xrightarrow{\text{Gradient Flow}} \mathcal{V}(\Psi) \\ \xrightarrow{\text{Spacetime Foliation}} (M, g) \xrightarrow{\text{Carrier Stabilizers}} G \xrightarrow{\text{Matter Duality}} T_{\mu\nu}. \end{array}$$

The Gauss–Codazzi–Ricci curvature decomposition establishes the four-block Hessian and canonical polar decomposition on carrier support. The autonomous gradient flow on configuration space proves local well-posedness and causal cone saturation. Frobenius integrability of the response flow along the distinguished shape line reconstructs four-dimensional spacetime and the Arnowitt–Deser–Misner initial-data constraints. Carrier bundle stabilizers determine the exact gauge group $[\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)]/\mathbb{Z}_6$. Variational spatial slice integrations determine matter couplings and quadratic gravitational attraction. This deductive progression establishes that the entire physical field theory is mathematically closed without external spacetime postulates.

H.2. Recurrent Trajectory Calibration and Monotonic Lyapunov Dissipation. Consider an integral curve $\tau \mapsto \Psi(\tau)$ of the autonomous response vector field on the quotient space. For recurrent configuration orbits that approximately return to earlier states, the return map evaluates character phases of the form $\chi(t) = a(t) \exp(i(q\phi(t) + vL(t)))$. In conjugate frequency space, the causal dispersion constraint takes the hyperbolic form $q^2 - \varepsilon^{-2}v^2 \geq 0$.

On the null threshold where contact equilibrium saturates, setting the scale $\varepsilon = 1$ yields the forward null branch condition

$$(185) \quad q = -v \implies \Delta\theta = q\Delta\phi + v\Delta L = 0.$$

Under this calibration condition, the total phase advance along null rays is stationary, $\Delta\theta = q\Delta\phi + v\Delta L = 0$. This stationarity selects coherent, non-dispersive wave packets that propagate along null geodesics at the speed of light.

Simultaneously, the state overlap amplitude $a(t)$ measures quantum dissipation along the trajectory. Differentiating with respect to the trajectory parameter yields the strictly negative Lyapunov decay rate

$$(186) \quad \frac{d}{dt} \ln a(t) = -\frac{1}{2} \|\mathcal{V}(\Psi(t))\|_G^2 \leq 0 \implies 0 < a(t) \leq a(0).$$

Because $\|\mathcal{V}(\Psi(t))\|_G^2 \geq 0$, the amplitude $a(t)$ decays monotonically toward the metric equilibrium minimum, establishing thermodynamic irreversibility and the arrow of time directly from configuration flow.

H.3. Mandelstam Area Derivative, Conserved Currents, and Charge Quantization. Gauge observables are represented gauge-invariantly on the loop space $\mathcal{L}(M) = C^\infty(S^1, M)$ by Wilson loops $W(\gamma) = \text{tr } \mathcal{P} \exp(i \oint_\gamma A)$. Deforming a loop γ by an infinitesimal loop bounding an area element $\delta\sigma^{\mu\nu}$ at event x , Stokes' theorem evaluates the connection holonomy. The functional derivative on loop space is the Mandelstam area derivative

$$(187) \quad \frac{\delta W(\gamma)}{\delta\sigma^{\mu\nu}(x)} = iF_{\mu\nu}(x)W(\gamma).$$

This formula extracts the local gauge curvature tensor directly from loop holonomies without requiring coordinate-dependent connection potentials.

Contracting the second Bianchi identity $\nabla_{[\mu} F_{\nu\rho]} = 0$ with the Lorentzian metric tensor derives the geometric current conservation law

$$(188) \quad \nabla^\mu J_\mu = 0 \iff d * F = *J.$$

Because the gauge field strength is sourced by interface matter currents via $D_X F = -I_X$, the geometric current J_μ is strictly conserved without introducing independent matter conservation postulates.

Topological charge quantization is geometrically required by the compact $U(1)$ topology of the gauge holonomy. Evaluating the holonomy around a spatial sphere surrounding a localized defect forces the phase to be an integer multiple of 2π , reproducing the Dirac quantization condition

$$(189) \quad eg = 2\pi n, \quad \nabla^\mu T_{\mu\nu} = 0 \quad \forall n \in \mathbb{Z}.$$

Furthermore, the contracted total Bianchi identity enforces divergence-free total stress-energy, $\nabla^\mu T_{\mu\nu}^{\text{total}} = 0$, across both regular strata and singular Whitney defect strata $\partial\Sigma_k$, completing the dynamical conservation laws.

APPENDIX I. OPERATIONAL FRAME RECONSTRUCTION AND DETERMINISTIC RECOVERY

Operational observers access physical spacetime and gauge fields strictly through laboratory measurement protocols. Orthonormal tetrad frames transported via Fermi–Walker transport [38, 123] eliminate non-inertial coordinate artifacts along timelike observer trajectories, while radar distance coordinates establish operational metric synchronization. The observation Jacobi operator [65] maps metric and connection variations to radar distances and Wilson loop interferometric phase shifts. On compact observation domains, bounded condition numbers ensure deterministic field recovery with controlled error amplification. Across singular Whitney defect strata [130, 129, 47], the observation operator remains Lipschitz continuous, while quantum measurement backreaction establishes an operational geometric ultraviolet cutoff $\ell_0 = \sqrt{G\hbar/c^3}$.

I.1. Operational Tetrad Frames and Radar Distance Synchronization. Let $\gamma(\tau)$ denote a smooth timelike observer trajectory parameterized by proper time τ , with future-directed four-velocity $\mathbf{u} = \dot{\gamma}$ satisfying $g(\mathbf{u}, \mathbf{u}) = -1$. An operational observer frame is an orthonormal tetrad $e_a = (e_0, e_1, e_2, e_3)$ where $e_0 = \mathbf{u}$ and $g(e_a, e_b) = \eta_{ab}$.

Spatial frame legs e_i ($i = 1, 2, 3$) are propagated along the worldline via Fermi–Walker transport [38, 123]

$$(190) \quad \nabla_{\mathbf{u}}^{\text{FW}} e_a = \nabla_{\mathbf{u}} e_a + g(e_a, \mathbf{a})\mathbf{u} - g(e_a, \mathbf{u})\mathbf{a} = 0.$$

where $\mathbf{a} = \nabla_{\mathbf{u}}\mathbf{u}$ is the four-acceleration. Fermi–Walker transport eliminates spurious coordinate rotations, providing standard inertial spatial axes for laboratory apparatus.

Radar coordinates are constructed by emitting null geodesics from the worldline at proper time $\tau_{(1)}$ to a target event p and detecting reflected signals at $\tau_{(2)}$. The operational radar distance is defined by

$$(191) \quad s_R(p) = \frac{1}{2}c(\tau_{(2)} - \tau_{(1)}).$$

In a normal neighborhood of the worldline, radar coordinates define a smooth chart in which the spacetime metric takes the Minkowski form $g_{\mu\nu}(0) = \eta_{\mu\nu}$ with corrections of order $\mathcal{O}(\|x\|^2)$ governed by the Riemann curvature tensor.

I.2. Deterministic Metric Recovery and Bounded Condition Numbers.

Field configurations are reconstructed from laboratory data through the observation Jacobi operator $\mathcal{J}$. Let $\delta(g, A)$ denote a perturbation of the metric tensor and gauge connection modulo gauge transformations, and let $\delta\mathcal{O} = (\delta s_R, \delta W(\gamma))$ denote the resulting variations in radar distance and Wilson loop holonomy phases

$$(192) \quad \mathcal{J}: \delta(g, A) \mapsto (\delta s_R, \delta W(\gamma)).$$

On compact measurement domains with non-degenerate probe distributions, the linear operator $\mathcal{J}$ is injective and possesses a closed range in the Hilbert space of observation data.

The numerical stability of the field reconstruction is governed by the observation condition number $\kappa(\mathcal{J}) = \|\mathcal{J}\| \|\mathcal{J}^{-1}\|$. Evaluating the operator norms on Sobolev spaces H^1 yields the uniform bound

$$(193) \quad \kappa(\mathcal{J}) = \|\mathcal{J}\| \|\mathcal{J}^{-1}\| \leq C_1 C_2 < \infty.$$

Because the condition number is strictly finite, errors in empirical radar and holonomy measurements are amplified by at most a finite factor $C_1 C_2$ during metric inversion. Field configurations are therefore deterministically recoverable from operational laboratory data.

I.3. Interface Lipschitz Stability and the Universal Geometric UV Cutoff.

Across singular Whitney defect strata $\partial\Sigma_k$, connection derivatives undergo jump discontinuities $\Delta S' = 2I_2$ and $\Delta J = 2J_0$. Despite these derivative discontinuities, the integrated observation operator satisfies a uniform Lipschitz continuity estimate

$$(194) \quad \|\mathcal{J}(\Psi_1) - \mathcal{J}(\Psi_2)\| \leq L_\Sigma \|\Psi_1 - \Psi_2\|.$$

where $L_\Sigma < \infty$ is an interface Lipschitz constant. Radar distance and Wilson holonomies integrate across singular defect strata continuously, ensuring that observational apparatus detects no singular coordinate artifacts.

At microscopic scales, localized quantum probes induce gravitational backreaction. Localizing a probe within spatial uncertainty Δx requires momentum fluctuation $\Delta p \geq \hbar/(2\Delta x)$, which concentrates energy and induces a metric disturbance $\Delta g \sim G\Delta p/c^3$. Operational measurability requires the measurement resolution

to exceed the induced metric distortion, $\Delta x \geq \Delta g$. This inequality enforces the universal geometric ultraviolet cutoff

$$(195) \quad \Delta x \geq \ell_0 = \sqrt{\frac{G\hbar}{c^3}} > 0.$$

The Planck scale ℓ_0 is therefore the minimal operationally resolvable distance in nature. Point-like coordinate singularities are unphysical idealizations excluded by the operational measurement protocol.

APPENDIX J. DETAILED FIELD MODELS, ZERO-MODE ELIMINATION, AND THE MCKEAN MASS GAP

Concrete field models realize metric equilibrium solutions across curved spatial geometries. Spherically symmetric configurations of coupled Einstein–Yang–Mills theory generate the non-singular Bartnik–McKinnon solitons [7] and colored black holes with non-Abelian hair. Zero-mode obstructions that affect flat tori are rigorously eliminated. Harmonic connection modes vanish topologically on simply connected leaves by Hodge’s theorem [60, 30, 124], while the radial dilatation mode is lifted into a positive-stiffness oscillation by the unilateral geometric scale-floor constraint $\varepsilon \geq b_3$ [70, 76]. Negative sectional curvature $K = -b_4^2$ activates McKean’s spectral theorem [91] to prove an exact non-perturbative Yang–Mills mass gap $m = |b_4|/2 > 0$, enforcing exponential clustering of correlation functions and completing the constitutive parameter system.

J.1. Non-Abelian Solitons and Colored Black Holes on Non-Flat Leaves.

Under static spherical symmetry, the spacetime metric and non-Abelian gauge connection admit the standard ansatz

$$(196) \quad ds^2 = -A(r)\mu(r)dt^2 + A(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2).$$

where $A(r) = 1 - 2m(r)/r$ is the metric potential and $w(r)$ is the Yang–Mills magnetic profile function. Substituting this ansatz into the coupled field equations yields a system of nonlinear ordinary differential equations for (m, μ, w) .

For the Bartnik–McKinnon soliton, imposing regularity at the origin $m(0) = 0$, $\mu(0) = 1$, $w(0) = 1$ and asymptotic flatness at infinity $m(r) \rightarrow M$, $w(r) \rightarrow \pm 1$ defines a shooting boundary value problem. Analysis of the phase-plane trajectories proves the existence of a discrete countable family of smooth, non-singular, particle-like soliton solutions indexed by the node count $n \in \mathbb{N}$.

For colored black holes, the metric potential vanishes at an event horizon radius $r_h > 0$, with $A(r_h) = 0$. The gauge profile satisfies $|w(r_h)| < 1$, establishing that non-Abelian hair persists outside the event horizon. These solutions confirm that curved spatial leaves support non-trivial, regular metric equilibrium configurations.

J.2. Topological Hodge Elimination and Unilateral Variational Dilatation

Lifting. Flat toroidal models suffer from two zero-mode obstructions, namely harmonic connection modes and the radial dilatation mode. Both obstructions are eliminated on non-flat spatial leaves.

On a compact simply connected three-dimensional Riemannian leaf Σ , such as the 3-sphere S^3 , Hodge’s theorem [60, 30, 124] establishes an isomorphism between the space of harmonic 1-forms and the first de Rham cohomology group

$$(197) \quad \mathcal{H}^1(S^3) = \{\alpha \in \Omega^1(S^3) : \Delta_H \alpha = 0\} \cong H_{\text{dR}}^1(S^3) = \{0\}.$$

Because the first Betti number vanishes, $b_1(S^3) = 0$, there are no non-trivial harmonic 1-forms. Every divergence-free connection variation is strictly co-exact, completely eliminating the harmonic zero mode from the physical variation space.

The radial dilatation mode $v_{\text{rad}} = (0, K_0)$ corresponds to global scale transformations along which the unconstrained curvature potential is invariant. Sourced matter introduces the unilateral scale-floor constraint [70, 76] $\varepsilon(x) \geq b_3 > 0$. The augmented Lagrangian functional is

$$(198) \quad \mathcal{L}(\Psi, \nu) = \Phi(\Psi) + \int_{\Sigma} \nu(x)(b_3 - \varepsilon(x)) d\mu_h.$$

where $\nu(x) \geq 0$ is the reactive normal strain density. At the saturated interface $\varepsilon = b_3$, contact equilibrium forces $\nu > 0$. Evaluating the second variation on the dilatation mode yields

$$(199) \quad D^2\mathcal{L}(v_r, v_r) = \nu b_3^2 > 0 \quad \text{for } \nu > 0.$$

The constraint term provides strictly positive stiffness, lifting the flat dilatation zero mode into the strictly positive discrete spectrum.

J.3. Geometric McKean Mass Gap Theorem and Constitutive Parameter Relations. Non-Abelian gauge energy curves the normal bundle sector into constant negative sectional curvature $K = -b_4^2 < 0$. Let $-\Delta$ denote the Laplace–Beltrami operator on a complete Riemannian manifold with sectional curvature bounded above by $-b_4^2$. By McKean’s spectral theorem [91], the bottom of the L^2 spectrum satisfies

$$(200) \quad \lambda_0 = \inf \text{spec}(-\Delta) \geq \frac{|K|}{4} = \frac{b_4^2}{4} > 0.$$

The physical wave operator is strictly bounded below by this spectral bottom. This establishes the exact, parameter-free Yang–Mills mass gap

$$(201) \quad m = \frac{|b_4|}{2} > 0 \implies |\langle \mathcal{O}(x) \mathcal{O}(0) \rangle| \leq C e^{-m|x|}.$$

This positive spectral gap enforces exponential clustering of vacuum correlation functions, proving that gauge excitations are massive and confined.

The physical parameters and coupling constants assemble into an invariant constitutive parameter system

$$(202) \quad (c, G, \hbar, \alpha, m, \ell_0) = \left(c, \frac{1}{16\pi b_4^2}, \hbar, \frac{e^2}{4\pi\varepsilon_0\hbar c}, \frac{|b_4|}{2}, \sqrt{\frac{G\hbar}{c^3}} \right).$$

This system connects the speed of light, effective gravitational constant, Planck constant, fine-structure constant, non-perturbative Yang–Mills mass gap, and minimal measurement scale into an exact system of constitutive parameter relations, concluding the formal development of the metric equilibrium theory.

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UNIVERSITY OF NORTH TEXAS

Email address: `sinamontazeri@my.unt.edu`